6DATA007W – Final Project Report

# An Empirical Analysis of Multidimensional Academic Stress Among Undergraduates Using PCA-Based Feature Extraction and Unsupervised Machine Learning

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**Use of Generative AI**

**Declaration**

This report has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

Generative artificial intelligence tools (ChatGPT) were used in a supportive capacity throughout this project. These tools were primarily used to explain programming code, improve code efficiency, and assist in debugging and error resolution. In addition, they were used to help identify relevant academic literature and to clarify complex theoretical concepts that were difficult to understand. All outputs generated were carefully reviewed, verified using academic sources, and refined to ensure accuracy and alignment with my own understanding. The final work reflects my independent analysis, interpretation, and academic judgement.

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**Abstract**

Academic stress has become a significant concern among university students, affecting both academic performance and overall well-being. This study presents a comprehensive data-driven analysis of academic stress among undergraduate students in the Faculty of Management and Finance at the University of Colombo. Addressing limitations in existing research, the study integrates statistical analysis, dimensionality reduction, clustering, and predictive modelling to capture the multidimensional nature of academic stress.

Data were collected from 1,763 students using a structured questionnaire based on the Perception of Academic Stress (PAS) scale. Following data pre-processing and reliability validation, descriptive and inferential analyses were conducted to examine stress patterns across demographic groups. Significant differences in stress levels were observed across gender, academic year, and degree programme, with stress generally increasing in higher academic years.

Principal Component Analysis (PCA) with Varimax rotation was applied to reduce dimensionality and identify underlying stress structures. Four key dimensions were extracted: academic workload and examination pressure (F1), academic and future concerns (F2), self-confidence and personal management (F3), and lecturer-induced pressure (F4). These dimensions explained 51.2% of the total variance, highlighting the complex and multidimensional nature of academic stress.

Using the PCA-derived factor scores, clustering techniques were applied to segment students into distinct stress profiles. A three-cluster solution using K-Means provided meaningful and interpretable groupings, enabling the identification of students with varying stress levels and characteristics.

As a secondary study, predictive modelling was conducted using demographic variables to estimate the identified stress dimensions. Variables with severe class imbalance were excluded to enhance model reliability. A multi-output regression approach was implemented, and results indicated that simpler models demonstrated better generalisation performance compared to more complex alternatives.

The findings reveal important insights, including lower self-confidence among first-year students and increasing stress levels in later academic years. These results emphasise the

importance of targeted interventions, such as confidence-building programmes and academic support mechanisms. Overall, this study contributes to both academic research and institutional decision-making by providing a structured understanding of academic stress and enabling early identification of at-risk students for improved well-being and academic outcomes.

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# 1. Introduction

This study analyses academic stress among management undergraduates using an adapted PAS scale and advanced methods to identify key stress dimensions and student segmentation.

## 1.1 Problem statement

Academic stress among undergraduates is a significant concern driven by both academic and non-academic factors, including socio-cultural, environmental, and psychological influences [1]. Among these, academic-related factors such as workload, examinations, and performance expectations have been consistently identified as primary sources of stress [2]. The transition to university intensifies this experience through greater academic demands, independence, and expectations [3].

Excessive academic stress is associated with a range of adverse outcomes, including physiological, cognitive, emotional, and behavioural effects [4]. It is also linked to poor academic performance, creating a cyclical relationship where stress both influences and is influenced by academic outcomes [4], [5]. Empirical evidence across diverse contexts confirms that academic stress is widespread and significantly affects student well-being and educational attainment [6]–[11]. Large-scale studies show that a substantial proportion of university students identify academic demands as their primary source of stress [12].

Within the Sri Lankan context, undergraduate students have been reported to experience moderate to high levels of perceived stress [13]. Additionally, a significant proportion of medical students (69.2%) experience psychological distress, including depression, anxiety, and stress, with 23% experiencing all three simultaneously [14]. These findings highlight the severity of academic stress and its psychological implications in the Sri Lankan higher education context.

Despite the growing body of research, limited attention has been given to management undergraduates within the Faculty of Management and Finance at the University of Colombo (UoC FMF). Preliminary informal discussions with students across all academic years further indicate the presence of academic stress.

The faculty comprises a linguistically and culturally diverse student population drawn from across the country. Students specialize from Year 3 onward. Sinhala and Tamil serve as the primary languages, while English is the medium of instruction, creating a learning environment that contributes to unique and unexplored academic stress dynamics.

Previous studies have examined academic stress across dimensions along with demographic differences like gender and academic year [8], [9]. However, most rely on descriptive or univariate methods, which do not fully capture its multidimensional nature.

The Perception of Academic Stress Scale (PAS) [9] has been developed to measure academic stress across multiple dimensions. However, limited attention has been given to its adaptation within the Sri Lankan context. Therefore, this study employs an adapted version of the PAS tailored to the UoC FMF.

Furthermore, there is a lack of studies integrating advanced multivariate and machine learning techniques in the analysis of academic stress. To address this gap, this study first applies descriptive and inferential statistical methods to examine differences in stress levels across student groups. PCA with Varimax rotation is then used to extract underlying stress dimensions specific to the study context. These context-specific dimensions are used in cluster analysis to segment students based on their stress profiles. Additionally, as a secondary study, predictive modelling is employed to estimate stress dimensions using selected demographic factors.

Therefore, this study conducts a comprehensive data-driven analysis of academic stress among undergraduates in UoC FMF by integrating a contextually adapted PAS scale with advanced statistical, multivariate, and clustering techniques. The findings enable the identification of distinct student segments based on stress profiles and the extraction of unique, context-specific academic stress dimensions. This provides deeper insights into stress patterns and supports the development of targeted interventions to address and minimise these issues, ultimately improving student well-being and academic outcomes.

## 1.2 Aims and Objectives

### Aim

The primary aim of this study is to conduct a data-driven analysis of academic stress among undergraduates in UoC FMF. It identifies underlying stress dimensions and segments students into distinct profiles using multivariate and machine learning techniques, including PCA with Varimax rotation and clustering. A secondary study aims to develop a predictive model using demographic data to predict academic dimensional stress and generate insights that support evidence-based decision-making and targeted interventions to improve student well-being and academic outcomes.

### Objectives

To achieve the above aim, the study is guided by the following objectives:

- To adopt the Perception of Academic Stress (PAS) scale as a validated instrument for measuring academic stress.
- To collect data from undergraduate students across all academic years using the questionnaire.
- To pre-process and clean the dataset by handling missing values and preparing it for analysis.
- To assess the internal consistency of the measurement instrument using Cronbach's Alpha.
- To perform descriptive analysis to summarise the demographic characteristics of the sample.
- To conduct inferential statistical analysis, including Welch's t-test and ANOVA, to examine differences in stress levels across gender, degree programme and academic year.
- To assess the suitability of the dataset for PCA using the KMO measure and Bartlett's Test of Sphericity.
- To apply Principal Component Analysis to reduce dimensionality and identify underlying latent stress factors.
- To apply Varimax rotation to enhance the interpretability of the extracted components and derive meaningful stress dimensions.

- To compute factor scores representing each stress dimension for individual students.
- To apply K-means and hierarchical clustering techniques to segment students based on the derived stress dimensions.
- To determine the optimal number of clusters for each method using validation techniques.
- To compare clustering performance between K-means and hierarchical clustering for the selected number of clusters using cluster validity metrics.
- To select the most appropriate clustering method based on superior validation performance.
- To perform cluster profiling on the final selected clustering solution to interpret and characterise student segments.
- Offer practical, evidence-based recommendations to assist UoC FMF in identifying and supporting students who are most at risk.

**Sub-Objectives**

- To develop predictive models to estimate pre-identified stress dimensions using demographic variables.
- To evaluate the performance of the predictive models using appropriate metrics.

# 2. Background

This section reviews existing literature related to academic stress among university students, with a focus on measurement scales, empirical findings, and analytical methods. It also examines the techniques and tools used in previous studies to analyse academic stress. The review highlights key limitations in existing approaches and provides a foundation for the methodological framework adopted in this study.

## 2.1 Literature review

### i) Measurement of Academic Stress (PAS Scale)

The Perception of Academic Stress (PAS) scale is widely used to measure students' perceptions of academic stress and its sources [9]. The scale was developed based on an extensive literature review that identified key dimensions of academic stress, including academic expectations, workload and examinations, and academic self-perceptions [9]. It is an 18-item self-report questionnaire using a five-point Likert scale to measure external stressors and internal psychological factors [9].

The PAS scale was refined through expert validation and pilot testing, ensuring clarity and reliability, and is now widely used to assess academic stress among university students.

### ii) Validation of the PAS Scale

The Perception of Academic Stress (PAS) scale has been validated across multiple contexts using established psychometric techniques. A large-scale study (5,499 students) identified a revised 16-item, three-factor model that demonstrated a better model fit while maintaining high reliability across gender groups [19].

Moreover, the scale has been empirically validated across multiple cultural and contextual settings. A study conducted in Pakistan translated and adapted the PAS into Urdu, supporting its applicability in a different cultural setting [20].

These findings collectively indicate that the PAS scale is both reliable and adaptable across diverse populations.

### iii) Empirical Findings on Academic Stress

Academic stress has been widely studied due to its impact on student well-being and performance. Studies have shown that high levels of academic stress are associated with increased anxiety and reduced mental well-being [15]. Similarly, academic stress negatively affects life satisfaction, particularly due

to performance-related pressures, while positive academic self-perception contributes to improved well-being [3].

In terms of academic outcomes, stress arising from workload, assessments, and work–life imbalance has been shown to negatively impact student performance, although the strength of this relationship varies [16]. Additionally, coping strategies such as planning and emotional support can mitigate these effects.

Furthermore, prior research indicates that academic stress is a multidimensional construct influenced by factors such as workload, time constraints, and performance pressure [9]. While many studies have examined relationships between stress and outcomes, some have applied multivariate techniques such as Principal Component Analysis (PCA) and Factor Analysis to better understand the underlying structure of academic stress by identifying latent dimensions, highlighting its complexity across different contexts [18], [20].

**iv) Methodological Approaches in Academic Stress Research**

Traditional research on academic stress has largely relied on statistical techniques to examine relationships and group differences. While these methods provide valuable insights, they are limited in their ability to capture the context's unique multidimensional nature of academic stress and the complex interrelationships among variables [3], [17].

To address this limitation, dimensionality reduction techniques such as Principal Component Analysis (PCA) have been increasingly applied to identify underlying stress factors. PCA reduces a large set of correlated variables into a smaller number of uncorrelated components, thereby simplifying complex datasets while retaining most of the original information [9]. However, in many studies, PCA is primarily used for validation or data reduction rather than subsequently clustering students based on the derived components [9], [19].

To enhance interpretability, rotation techniques such as Varimax are often applied. Varimax rotation simplifies factor structures by maximizing the variance of factor loadings, allowing each variable to load strongly onto a single factor and facilitating clearer identification of latent stress dimensions [9]. Despite these, the combined use of PCA and rotation techniques to derive well-defined, interpretable stress dimensions along with clustering appears relatively limited in academic stress research. Building on the stress dimensions extracted through PCA, this study further employs cluster analysis to group students into distinct segments, thereby identifying different stress profiles among students based on these underlying dimensions.

Previous studies have demonstrated the effectiveness of multi-output XGBoost models in capturing complex, non-linear relationships and improving predictive performance compared to traditional

models. For instance, a recent study showed that multi-output XGBoost outperformed classical approaches by simultaneously predicting multiple related outcomes and capturing interactions among variables [28]. However, such applications have primarily focused on non-academic contexts. Therefore, this study extends the use of multi-output XGBoost to predict multidimensional academic stress, addressing a gap in existing research.

Beyond methodological considerations, several limitations are evident in existing studies. Many rely on small or non-representative samples, limiting the generalizability of findings [16], [17]. Demographic imbalances, particularly in gender distribution, are also frequently reported, which may affect the reliability of comparisons [16], [18]. Furthermore, research often emphasizes limited demographic comparisons, with less attention given to variations across academic years, particularly in the Sri Lankan university context.

From a measurement perspective, studies rely on general stress assessment tools such as the Perceived Stress Scale (PSS), which may not fully capture stressors specific to academic environments. As a result, factors such as workload, examinations, and peer competition may be underrepresented, highlighting the importance of academic stress-specific instruments such as the PAS scale [22].

Overall, the literature suggests that while a range of statistical and analytical techniques have been applied in academic stress research, the integrated use of dimensionality reduction methods such as PCA, clustering techniques for student segmentation, remains relatively limited, particularly in context-specific settings.

*Table 1: Summary of Selected Literature on Academic Stress*

| Ref | Dataset | Scale / Instrument | Methodology | Relevance | Research Gap |
|---|---|---|---|---|---|
| [9] | 100 students (+12 experts) | PAS | PAS Scale development, reliability, PCA and Varimax. | Establishes PAS and key stress dimensions; foundation for measurement and PCA | Small sample; No advanced analysis (clustering/ML) after PCA |
| [15] | 843 college students | PAS, SWEMWBS | Survey (SWEMWBS & PAS); correlation, MANOVA, and chi-square analysis. | Supports PAS scale; links stress with well-being and demographic differences. | Limited generalizability due to sample bias; correlational data; lacks diverse representation and advanced modelling. |
| [16] | 256 students | SISCO | Mixed-method; descriptive, t-test, correlation | Identifies multidimensional stress factors and coping strategies | Small sample; no clustering or ML |
| [17] | 150 students | PAS | Survey; descriptive & gender analysis | Sri Lankan context supports demographic analysis | Traditional methods; no PCA, clustering, or ML |
| [18] | 339 students | PAS | Reliability, validity, EFA | Identifies stress dimensions; supports PCA use | No predictive or clustering methods; limited scope |
| [19] | 5,499 students | PAS | EFA, PCA, CFA | Strong validation of PAS supports dimensionality reduction | No clustering or ML; validation-focused |
| [22] | 444 students | PSS | PCA & ML | Combines dimensionality reduction and prediction | No clustering; limited generalizability |
| [23] | 1375 students | EUDIQ-R, USEI | CFA, reliability, invariance | Supports psychometric validation and structure | Not stress-focused; no clustering or ML |
| [24] | 452 students | Behavioral data | Hierarchical clustering, PCA, Silhouette | Supports clustering and segmentation | Not stress-based; no predictive modelling |
| [26] | 1,100 students | Dataset features | ML (LR, RF, XGBoost) | Supports stress prediction using ML | No validated scales; no PCA or clustering |
| [25] | 1,000 records | Dataset variables | K-means clustering | Supports the segmentation approach | No validated scales; limited variables |
| [27] | 54 students | PSS, GAD-7 | ML | Supports ML prediction and feature importance | Small sample; not academic stress |
| [28] | 147 soil samples | N/A | Multi-output XGBoost, SHAP | A multi-output XGBoost model to predict Phosphorus adsorption at multiple levels at once. | Not academic stress-related; domain-specific |

## 2.2 Review of Methods and Applications

### 2.2.1 Statistical Methods for Academic Stress Analysis

Cronbach's Alpha is widely used to assess the internal consistency of measurement instruments such as the Perception of Academic Stress (PAS) scale [9], [17], with values above 0.7 indicating acceptable reliability. It ensures consistency but does not capture the underlying dimensional structure of the data.

Descriptive statistics are commonly used to summarise dataset characteristics and identify general patterns in academic stress levels [17]. It provides a clear overview of data distribution, but does not establish relationships or statistical significance.

Inferential statistical techniques, including Welch's t-test and Analysis of Variance (ANOVA), are frequently applied to examine differences in stress levels across demographic groups [17], [21]. Welch's t-test is suitable for comparing two groups without assuming equal variances, while ANOVA enables comparisons across multiple groups. Despite their effectiveness in hypothesis testing, these methods are limited in capturing complex relationships and multidimensional structures within the data.

### 2.2.2 Dimensionality Reduction Techniques

*i)* Principal Component Analysis (PCA)

PCA is widely used in academic stress research to reduce dimensionality and identify latent structures [9], [19]. It transforms correlated variables into a smaller set of uncorrelated components while retaining most of the dataset's variance, making it suitable for analysing multidimensional constructs such as academic stress. Prior studies have applied PCA to extract key stress dimensions from survey data, improving analytical efficiency and interpretability [9], [19].

The suitability of PCA is assessed using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity, which evaluate sampling adequacy and variable correlation [9], [18]. Components are selected based on eigenvalues, with scree plots used to determine the optimal number. PCA effectively simplifies complex datasets and highlights key stress dimensions, but interpretation remains subjective, as components must be meaningfully labelled by the

researcher. Furthermore, PCA does not capture relationships between variables or differences across groups.

*ii)* Varimax Rotation

Varimax rotation is applied after PCA to enhance interpretability [9]. This orthogonal rotation method maximizes the variance of factor loadings, producing a simpler structure in which each variable loads strongly onto a single component [9]. This enhances clarity in identifying and labelling stress dimensions, making results more meaningful, though interpretation depends on the researcher and may vary across studies.

### 2.2.3 Clustering Techniques

While PCA identifies underlying stress dimensions, it does not capture heterogeneity among students. Clustering techniques address this limitation by grouping individuals into distinct profiles based on similarities in their characteristics, making them a key unsupervised learning approach in educational data analysis.

K-means and hierarchical clustering are widely used in student-related research. K-means partitions observations into predefined clusters by minimizing within-cluster variance and is valued for its simplicity, efficiency, and scalability. It has been applied to group students based on behavioural and academic characteristics, enabling identification of meaningful segments [25]. Hierarchical clustering constructs a dendrogram representing nested groupings and does not require a predefined number of clusters, making it suitable for exploratory analysis and understanding relationships among student behaviours [24].

K-means requires a predefined number of clusters, which may lead to suboptimal results, while hierarchical clustering is computationally intensive and less scalable. Utilizing both improves robustness, and validation methods like the Elbow method, Silhouette score, and Calinski–Harabasz index help assess clustering quality and determine optimal structures.

Clustering enables the identification of hidden patterns and student segments, supporting targeted interventions and data-driven decision-making. However, results are sensitive to parameter selection and require careful interpretation.

### 2.2.4 Secondary Study-Predictive Modeling

Traditional regression models are limited in capturing complex relationships; therefore, advanced machine learning methods such as XGBoost are increasingly used for their ability to model non-linear interactions and improve predictive accuracy. As an ensemble method based on gradient boosting, XGBoost effectively captures complex patterns in data. Prior studies demonstrate that machine learning approaches, including XGBoost, can accurately predict student stress and identify underlying patterns [26], [27].

Recent advancements extend gradient boosting to multi-output settings, enabling the simultaneous prediction of multiple interrelated outcomes. This is particularly relevant for academic stress, which is inherently multidimensional. Multi-output XGBoost models have shown strong performance in modelling such complex relationships, as evidenced in applications like soil adsorption modelling [28], supporting their suitability for predicting multidimensional stress profiles.

Model performance is commonly evaluated using metrics such as $R^2$, RMSE, and MAE to assess predictive accuracy. Predictive modelling supports early identification of at-risk students; however, its effectiveness depends on input data quality.

## 2.3 Review of Tools

This section reviews the tools and technologies relevant to data-driven model development for academic stress analysis. It evaluates their strengths, limitations, and suitability, while considering alternative tools. The selected tools support data pre-processing, statistical analysis, dimensionality reduction, clustering, and predictive modelling.

### 2.3.1 Programming Languages

Python is widely used in data analytics and machine learning due to its simplicity, flexibility, and extensive library ecosystem [29]. It supports rapid development and integration of statistical and machine learning techniques, making it suitable for academic research and predictive modelling.

R is a common alternative, particularly strong in statistical analysis and visualization. However, it is less flexible for integrating machine learning workflows and large-scale data processing.

Python's strengths include ease of use, strong community support, and a wide range of libraries, while its main limitation is slower execution compared to compiled languages like C++.

### 2.3.2 Data Processing and Statistical Libraries

Pandas and NumPy are widely used for data manipulation and numerical computation. Pandas provides efficient data structures for structured datasets, while NumPy enables fast numerical operations [30], [31]. They support key pre-processing tasks such as handling missing values, encoding variables, and transforming data, though they can be memory-intensive for large datasets.

For statistical analysis, SciPy and Statsmodels provide reliable implementations of techniques such as t-tests and ANOVA [34], [35]. While widely used for their robustness, they require a solid understanding of statistical assumptions.

Tools like SPSS and Stata offer user-friendly interfaces but lack flexibility for integration with machine learning workflows. Therefore, Python-based libraries provide a more scalable and adaptable solution.

### 2.3.3 Machine Learning Libraries

Scikit-learn is one of the most widely used machine learning libraries due to its simplicity, extensive documentation, and support for a wide range of algorithms [32]. It provides implementations for techniques like PCA, clustering, regression, and pre-processing functions including standardization and data splitting. These features make it suitable for developing end-to-end analytical pipelines

XGBoost is a specialised library for gradient boosting that is known for its high predictive accuracy and efficiency [33]. It is particularly effective in modelling complex and non-linear relationships, making it suitable for predictive modelling tasks. Compared to traditional models, XGBoost [28] achieves superior performance due to regularisation and optimisation techniques. However, it requires careful hyperparameter tuning and is less interpretable than simpler models.

The factor_analyzer library provides tools for factor analysis, including KMO and Bartlett's tests and Varimax rotation. While it enhances the ability to interpret latent structures, its outputs require careful interpretation, and it is less commonly used than mainstream libraries.

### 2.3.4 Data Visualisation Tools

Matplotlib and Seaborn are used for data visualization in academic research. Matplotlib provides flexible, detailed plotting, while Seaborn offers higher-level statistical visualizations [36], [37]. They support essential graphs such as distributions, heatmaps, and scree plots for exploratory analysis. Power BI offers advanced interactive visuals, but is less suitable for Python-based workflows. Although Matplotlib and Seaborn require effort for formatting, they provide greater control and flexibility for research.

### 2.3.5 Development Environment

Google Colab is a cloud-based development environment that provides Jupyter Notebook functionality, enabling interactive coding and model development [38]. It allows users to run code without local installation.

Alternative environments include Jupyter Notebook and Visual Studio Code. While Jupyter offers similar functionality, it requires local setup, and VS Code provides advanced development features but is less suited for exploratory analysis. Colab's strengths include accessibility, cloud execution, and ease of use, although it depends on internet connectivity and has runtime limitations.

### Conclusion of Tools

Overall, the selected tools, particularly Python and its associated libraries, provide a flexible and efficient framework for data processing, statistical analysis, and machine learning. Their integration supports scalable and accurate analysis of academic stress, while offering advantages over alternative tools in terms of adaptability, performance, and ease of implementation.

# 3. Legal, Ethical, Sustainability, and Other Considerations

## 3.1 Legal Considerations

When handling academic stress-related data within an educational context, strict adherence to data protection and privacy regulations is essential. In this study, no personally identifiable information (PII) was collected. All responses were fully anonymised, with no names, student identification numbers, or identifiable attributes recorded, ensuring that individual participants cannot be traced.

The study complies with the Personal Data Protection Act No. 9 of 2022 by adhering to key principles such as data minimisation, anonymisation, lawful data collection, and the protection of participant privacy. Formal permission to collect data was obtained from the relevant faculty, and the approval letter is included in Appendix B. Data integrity and security were maintained by storing the dataset in a controlled environment and restricting unauthorized access or modification.

## 3.2 Ethical Considerations

Participation in the survey was entirely voluntary, and respondents had the right to withdraw at any stage without any consequences. Refer to annexure B for the Survey Questionnaire (Google Form Link).

The questionnaire was designed using neutral and non-invasive language to minimise psychological discomfort and reduce response bias. Participants were clearly informed that the study was conducted solely for academic and supportive purposes, and not for diagnostic, evaluative, or decision-making use at an individual level.

All results were analysed and presented in aggregated form, ensuring that no individual participant could be identified, labelled, or adversely affected by the findings.

## 3.3 Data Quality and Responsible Data Use

Ensuring data quality was a key priority throughout the study. Data veracity was supported by a well-structured questionnaire, consistent response scales, and systematic data-cleaning procedures, including the removal of incomplete or invalid responses.

Reliability of the measurement instrument was assessed using Cronbach's Alpha to confirm internal consistency. In addition, all pre-processing and analytical steps were carefully documented to ensure transparency and reproducibility of results.

Responsible data use was maintained by interpreting results cautiously, avoiding misleading conclusions, and ensuring that findings were used only for academic and research purposes.

## 3.4 Software / Application Ethics

The analysis was conducted using Python and its associated libraries within the Google Colab environment. These tools were used responsibly to ensure accuracy, transparency, and reproducibility of the analysis. All analytical procedures were conducted using established methodologies.

## 3.5 Social and Sustainability Considerations

This study contributes to social sustainability by enhancing the understanding of academic stress and its impact on student well-being. By identifying context unique academic stress dimensions and enabling the prediction of stress levels from demographic characteristics, the findings can support early identification and targeted interventions to improve students' academic experiences and mental health.

The research has minimal environmental impact, as all processes were conducted digitally using online surveys and cloud-based tools, thereby reducing the use of physical resources.

Overall, the study provides socially responsible and data-driven insights for students, educators, and decision-makers, while ensuring that no automated decisions or individual-level judgments are made based on the analysis.

# 4. Methodology

This study adopts a data-driven approach to analyse academic stress among university students.

The methodology is structured into two main analytical components:
(1) Study 1 – Descriptive and inferential analysis, followed by Principal Component Analysis with Varimax rotation and Clustering
(2) Study 2 – Predictive modelling

The overall pipeline begins with data collection and pre-processing, followed by statistical analysis, dimensionality reduction, clustering, and predictive modelling.

## 4.1 Data

### 4.1.1 Data Collection

Data were collected through a structured online questionnaire distributed among undergraduate students of the UoC FMF. The survey captured the main dimensions of academic stress along with demographic information.

The questionnaire was developed based on the validated Perception of Academic Stress (PAS) scale to ensure reliability and content validity. It was distributed via Google Forms through WhatsApp groups after the end of lectures across all academic years.

Participation was voluntary and anonymous, with no personally identifiable information collected. The final dataset consisted of 1763 valid responses.

### 4.1.2 Dataset Structure

The dataset comprises categorical and ordinal variables organised into demographic information and academic stress measurement variables. Refer to Appendix A Table 06: Description of Demographic and Academic Stress Questions.

Demographic variables were selected based on the PAS framework and supporting literature [9], enabling analysis of stress variations across student groups. These include gender, academic

year, marital status, degree programme, academic failure, faculty learning environment, home environment, and age group.

Gender was included due to documented differences in stress perception across male and female students [17]. Academic year captures progression within undergraduate studies, allowing analysis of adjustment challenges in early years and future-related pressures in later years.

Marital status, included in the original PAS scale [9], accounts for potential differences in external responsibilities. The degree programme was used instead of faculty to provide a more context-specific analysis within the Faculty of Management and Finance, enabling identification of specialization-based stress patterns.

Academic failure was included as it is closely linked to examination-related stress [9], while faculty and home environment variables capture both institutional and external influences. Age was incorporated to reflect variability in student backgrounds due to differing university entry pathways.

The stress measurement variables consist of Likert-scale items grouped into three dimensions: academic expectations, workload and examinations, and academic self-perception. These variables provide a comprehensive framework for analysing academic stress and its influencing factors.

## 4.2 Data Pre-Processing

### 4.2.1 Data Cleaning

Data cleaning was performed to ensure consistency and usability of the dataset. The Timestamp column was removed as it was not relevant to the analysis. One fully empty record was identified and removed, while no further missing values were present due to mandatory questionnaire responses.

Column names were standardised by removing spaces and formatting inconsistencies, and survey questions were renamed into concise variable names to improve clarity and facilitate analysis. Refer to Appendix A for the renamed description of the questionnaire.

### 4.2.2 Reverse Coding of Questionnaire Items

Some questionnaire items were negatively worded to reduce response bias and discourage uniform or inattentive answering patterns.; therefore, to maintain consistency in the

measurement scale, where higher values indicate higher stress, these items were reverse-coded using the transformation. (Reversed Score = 6 − Original Score)

The reverse-coded variables are presented in Appendix A, Table 07: Reverse-Coded Variables.

### 4.2.3 Overall Stress Measure

An overall stress score was computed by taking the mean of all Likert-scale items for each respondent. This provided a continuous measure of academic stress, which was used in inferential analysis to examine differences across demographic groups.

## 4.3 Descriptive Analysis

Descriptive statistics were used to summarise dataset characteristics. Frequencies and percentages were applied to analyse demographic distributions and identify initial patterns in stress levels.

## 4.4 Inferential Analysis

Inferential statistical techniques were applied to examine differences in academic stress levels across selected demographic groups. An independent Welch's t-test was conducted to compare mean stress levels between male and female students, as it does not assume equal variances between groups.

A one-way Analysis of Variance (ANOVA) was used to assess differences in mean stress levels across academic years and degree programmes, as these variables consist of more than two categories. When statistically significant results were observed, Tukey's Honestly Significant Difference test was performed to identify specific group differences [21].

All statistical tests were conducted at a significance level of 0.05, where a p-value less than 0.05 indicates a statistically significant difference between groups. The following hypotheses were tested:

**Gender**

$H_0$: There is no significant difference in mean stress levels between male and female students.
$H_1$: There is a significant difference in mean stress levels between male and female students.

**Academic Year**

$H_0$: There is no significant difference in mean stress levels across academic years.

$H_1$: There is a significant difference in mean stress levels among at least two academic year groups.

**Degree Programme**

$H_0$: There is no significant difference in mean stress levels across degree programmes.

$H_1$: There is a significant difference in mean stress levels across at least two degree programmes.

Only the above demographics were included due to their relevance and suitability for statistical comparison. Variables with highly imbalanced distributions were excluded to ensure the reliability of results.

## 4.5 Dimensionality Reduction

To address high dimensionality, Principal Component Analysis was applied to transform correlated variables into a smaller set of uncorrelated components while retaining most of the variance. This enabled the identification of latent academic stress dimensions specific to the UoC FMF context, providing a structured understanding of stress patterns.

The suitability of the dataset for PCA was assessed using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity. A KMO value of 0.846 indicates good sampling adequacy, while a statistically significant Bartlett's Test ($p < 0.001$) confirms that the variables are sufficiently correlated [9].

Components were extracted based on eigenvalues, scree plot analysis, and cumulative variance explained. A four-factor solution was selected based on the Kaiser criterion and interpretability [9].

Varimax rotation was applied to improve clarity of factor loadings, allowing each variable to load strongly onto a single component. The resulting dimensions represent workload and examination pressure, academic stress and future concerns, academic self-confidence, and lecturer-induced pressure.

These factors explain approximately 51.2% of the total variance, indicating a strong representation of the underlying stress structure.

## 4.6 Clustering Analysis

### 4.6.1 Clustering Approach

Following the extraction of the four principal components (F1–F4), clustering techniques were applied to group students into distinct segments based on their academic stress profiles. The PCA-derived factor scores were used as inputs, as they reduce dimensionality and multicollinearity while preserving the most meaningful variation in the data. This approach enables the identification of homogeneous student groups and supports targeted interventions based on stress patterns.

4.6.2 Clustering Techniques

Two unsupervised learning methods were implemented: K-Means clustering and Hierarchical (Agglomerative) clustering. K-Means partitions observations into K clusters by minimising within-cluster variance and assigning each observation to the nearest centroid, making it efficient and suitable for large datasets. In contrast, hierarchical clustering follows a bottom-up approach, iteratively merging observations based on similarity using Ward's linkage method, allowing exploration of nested cluster structures.

4.6.3 Determination of Optimal Number of Clusters

The optimal number of clusters (K) was determined using multiple validation techniques, including the Elbow Method, Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. While the two-cluster solution showed slightly better statistical performance, the differences were marginal. Therefore, a three-cluster solution was selected due to its greater interpretability and its ability to capture varying levels of academic stress among students.

4.6.4 Model Evaluation and Selection

Both K-Means and hierarchical clustering for 3 clusters were evaluated using internal validation metrics to assess cluster quality. The results indicated that K-Means produced more distinct and well-separated clusters compared to hierarchical clustering. Based on this evaluation, K-Means was selected as the final clustering method due to its superior performance and clearer segmentation.

#### 4.6.5 Cluster Formation

Using the selected K-Means model, students were grouped into three clusters based on their stress profiles derived from the PCA components. Each observation was assigned a cluster label, representing different levels or patterns of academic stress.

#### 4.6.6 Cluster Profiling

To interpret the clusters, profiling was conducted using both stress dimensions and demographic characteristics. This involved analysing the mean values of F1–F4 and examining the distribution of demographic variables. This process enabled the identification of meaningful student segments and provided deeper insights into variations in academic stress across groups.

## 4.7 Predictive Modelling (Secondary Study)

#### 4.7.1 Objective and Variable Selection

Predictive modelling was conducted to estimate stress dimensions using selected demographic variables identified through descriptive analysis. Variables with severe class imbalance, such as marital status, were excluded to improve model reliability. PCA-derived factor scores (F1–F4) were used as target variables, enabling a structured and interpretable modelling approach.

#### 4.7.2 Data Preparation

Pre-processing steps were applied to address class imbalance and ensure consistency. Degree programmes were grouped to reduce sparsity, and categories with low representation were merged. Categorical variables were transformed using one-hot encoding, converting them into a numerical format suitable for machine learning models.

#### 4.7.3 Model Development

The dataset was split into training and testing sets using an 80:20 ratio to evaluate generalisation performance. A fixed random state ensured reproducibility.

XGBoost was used as the predictive model, implemented through a “MultiOutputRegressor” to handle multiple target variables simultaneously. Model parameters were configured to balance performance and complexity.

#### 4.7.4 Model Evaluation

Model performance was evaluated using $R^2$, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), providing a comprehensive assessment of predictive accuracy.

#### 4.7.5 Justification of Approach

The use of PCA-derived stress dimensions as targets provides a more interpretable framework compared to individual questionnaire items. Pre-processing steps improve model stability, while XGBoost enables modelling of complex, non-linear relationships. The train-test split ensures reliable evaluation on unseen data.

## 4.8 Summary of Methodological Approach

The methodology integrates statistical analysis, dimensionality reduction, and clustering to identify context-specific stress dimensions and student segments. It further incorporates predictive modelling to estimate student stress profiles, supporting data-driven decision-making in higher education.

# 5. Tools

Based on the evaluation in Section 2.3, the following tools were selected to implement the academic stress analysis model. These technologies support efficient functioning across the entire pipeline.

## 5.1 Programming Language

Python was used as the primary programming language to integrate data pre-processing, statistical analysis, and machine learning within a single workflow. Its simplicity, readability, and extensive ecosystem enabled efficient handling of survey data, implementation of statistical tests, and development of predictive models.

Compared to alternatives such as R, Python provides greater flexibility for machine learning implementation and end-to-end analytical workflows. Its widespread use also supports reproducibility, scalability, and strong community support.

## 5.2 Data Analysis and Machine Learning Libraries

Several Python libraries were applied across different stages of the analysis. Pandas and NumPy were used for pre-processing tasks, including data cleaning, transformation, encoding, and efficient handling of structured survey data. While highly efficient, they may become memory-intensive for large datasets.

Scikit-learn was used to implement PCA for dimensionality reduction, K-means and hierarchical clustering for segmentation, and regression models for predictive analysis. A multi-output regression approach was applied to predict stress dimensions (F1–F4) simultaneously. Scikit-learn also supported feature standardisation, train-test splitting, and model evaluation using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and $R^2$. Although effective for building analytical pipelines, it requires careful model selection and parameter tuning.

XGBoost was implemented to model complex non-linear relationships between demographic variables and stress dimensions. Its gradient boosting framework provides high predictive accuracy and robustness compared to traditional models, though it requires careful hyperparameter tuning and is less interpretable.

SciPy and Statsmodels were used for inferential statistical analysis, including one-way ANOVA and Tukey's HSD test to identify differences across student groups. These tools provide reliable statistical testing but require adherence to underlying assumptions.

The Factor Analyzer library was used to assess PCA suitability using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity, and to apply Varimax rotation for improved interpretability. Pingouin was used to compute Cronbach's Alpha for reliability assessment. These tools support robust statistical validation, though their outputs require careful interpretation.

## 5.3 Data Visualisation

Matplotlib was used to generate key visualisations, including scree plots and stress distribution graphs, supporting data exploration and interpretation. It offers flexible and customisable outputs suitable for academic reporting, although additional formatting may be required. Power BI was also utilised to create interactive dashboards, enhancing the presentation of findings and enabling dynamic exploration of patterns.

## 5.4 Development Environment

All analysis was conducted using Google Colab, a cloud-based Python environment with an interactive notebook interface and seamless Google Drive integration. It enabled efficient experimentation without local setup, supported automatic saving, and allowed easy accessibility across devices. However, it depends on internet connectivity and has runtime limitations.

# 6. Model development

This section outlines the development of the analytical framework used to examine academic stress among undergraduate students. It covers each stage of the process, including data pre-processing, reliability analysis, descriptive statistics, hypothesis testing, dimensionality reduction using PCA, and clustering techniques. The study was implemented using Python in the Google Colab environment, with the dataset stored and accessed via Google Drive to ensure an efficient and reproducible workflow.

## 6.1 Data Import and Preparation

The dataset used in this study was gathered through a survey and stored in the form of a CSV file on Google Drive.

```
from google.colab import drive
drive.mount('/content/drive')

Mounted at /content/drive

import pandas as pd
df = pd.read_csv('/content/drive/MyDrive/FYP_IIT/Uoc_FMF.csv', encoding='latin1')
df.head()
```

*Figure 1: Dataset Loading*

## 6.2 Data Pre-processing

Data pre-processing involved removing the timestamp variable, renaming columns for consistency, and cleaning unwanted characters. Missing values were handled to ensure data quality. Categorical variables were standardised, and reverse coding was applied to maintain consistent scale direction. Skewness was assessed to confirm the data's suitability for further analysis without transformation.

### 6.2.1 Removal of non-contributory columns

The timestamp column generated by Google Forms was removed as it was not relevant to the analysis.

```
df = df.drop(columns=["Timestamp"])
```

*Figure 2: Removal of timestamp*

### 6.2.2 Missing Value Handling and Column Cleaning

The dataset was examined for missing values, and a single incomplete response identified as a test entry was removed. Additionally, column names were cleaned by removing extra spaces and line breaks to ensure consistency and ease of analysis.

```
df = df.dropna(how='all')

df.columns = df.columns.str.strip()
df.columns = df.columns.str.replace("\n", "", regex=True)
```

*Figure 3: Handle missing values*

### 6.2.3 Renaming Columns

The original column names were simplified to improve clarity and ease of analysis. This ensured better readability and facilitated efficient data manipulation throughout the analysis.

```
df = df.rename(columns={
    'What is your gender?': 'gender',
    'What is your marital status?': 'marital_status',
    'Have you ever failed an academic course/module at university?': 'academic_failure',
    'How satisfied are you with your learning environment at the Faculty?': 'faculty_environment',
    'Are you satisfied with your home environment for studying?': 'home_environment',
    'What is your Age?': 'age_group',
    'What is your Academic Year?': 'academic_year',
    'What is your degree programme.': 'degree_program',
    'Competition with my peers for grades is quite intense': 'peer_competition',
    'I feel my lecturers are very strict and often point out mistakes in my work': 'lecturer_strictness',
    'My lecturers expect more from me than I can realistically do.': 'lecturer_expectations',
    'The unrealistic expectations of my parents stress me out.': 'parental_expectations',
    'The time allocated for lectures, laboratory sessions, and independent study is enough for me to manage my academic workload effectively.': 'time_management',
    'The size of the curriculum (workload) is excessive.': 'curriculum_load',
    'I believe that the amount of work assignments is too much.': 'assignment_load',
    'I am unable to catch up if I fall behind in my work.': 'difficulty_catching_up',
    'I have enough time to relax after completing my academic work.': 'relaxation_time',
    'The examination questions are usually difficult.': 'exam_difficulty',
    'The allocated examination duration is too short for me to complete my answers properly.': 'exam_time_pressure',
    'Examination periods are very stressful to me.': 'exam_stress',
    'I am confident that I will be a successful student.': 'academic_confidence',
    'I am confident that I will be successful in my future career.': 'career_confidence',
    'I find it easy to make academic decisions.': 'decision_making',
    'I fear failing courses/modules this year.': 'module_failure',
    'I think that my worry about examinations is weakness of character.': 'exam_concern',
    'Even if I pass my exams, I am worried about getting a job': 'employment_concern'
})
```

*Figure 4: Rename column names*

### 6.2.4 Reverse Coding

Reverse coding was applied to selected variables to ensure consistency in the direction of the Likert scale responses. Items where higher values did not represent higher stress were transformed using reverse coding, so that all variables aligned in the same direction.

```
reverse_items = [
    "peer_competition",
    "lecturer_strictness",
    "lecturer_expectations",
    "parental_expectations",
    "curriculum_load",
    "assignment_load",
    "difficulty_catching_up",
    "exam_difficulty",
    "exam_time_pressure",
    "exam_stress",
    "module_failure",
    "exam_concern",
    "employment_concern"
]

# Apply reverse coding (1-5 Likert scale)
for col in reverse_items:
    df[col] = 6 - df[col]
```

*Figure 5: Reverse coding*

### 6.2.5 Cleaning special characters

Selected columns were converted to string format, and special symbols and non-alphanumeric characters were removed. For the age group variable, only numeric values were retained, and formatting was standardised by inserting appropriate separators to maintain a consistent structure. These steps ensured that categorical data were clean, consistent, and suitable for further analysis.

```
# Columns to clean (remove weird symbols like ⊠)
cols_to_clean = [
    "marital_status",
    "academic_failure",
    "home_environment",
    "academic_year"
]
for col in cols_to_clean:
    df[col] = (
        df[col]
        .astype(str)
        .str.replace(r"[^\w\s]", "", regex=True)  # remove weird symbols
        .str.strip()
    )
df["age_group"] = (
    df["age_group"]
    .astype(str)
    .str.replace(r"[^\d]", "", regex=True)   # keep only numbers
    .str.replace(r"(\d{2})(\d{2})", r"\1-\2", regex=True)  # insert hyphen
)
```

*Figure 6: Cleaning special characters*

### 6.2.6 Skewness Assessment

An overall stress score was calculated by averaging all relevant survey items for each respondent, providing a single composite measure of academic stress. Skewness was then assessed to examine the distribution of the scores and determine whether the data were approximately symmetrical or skewed, ensuring its suitability for further statistical analysis.

```
all_items = [
    'peer_competition','lecturer_strictness','lecturer_expectations','parental_expectations',
    'time_management','curriculum_load','assignment_load','difficulty_catching_up',
    'relaxation_time','exam_difficulty','exam_time_pressure','exam_stress',
    'academic_confidence','career_confidence','decision_making',
    'module_failure','exam_concern','employment_concern'
]

df['overall_stress'] = df[all_items].mean(axis=1)
df['overall_stress'].skew()
```

*Figure 7: Skewness measurement*

### 6.2.7 Exporting Cleaned Dataset

The cleaned dataset was exported as a CSV file and uploaded to Google Drive to ensure accessibility and secure storage.

```
from google.colab import files

df.to_csv("UOC_FMF_Cleaned_Dataset.csv", index=False)
files.download("UOC_FMF_Cleaned_Dataset.csv")
```

*Figure 8: Exporting Cleaned Dataset*

## 6.3 Reliability Analysis using Cronbach's Alpha

Cronbach's Alpha was computed to assess the internal consistency of the survey items used to measure academic stress. The reliability coefficient was calculated using the *Pingouin* library. The resulting alpha value indicated the degree to which the items consistently measured the underlying construct, confirming the suitability of the data for further analysis. This was calculated overall and dimension-wise.

```
from pingouin import cronbach_alpha

all_items = academic_expectations + workload_examinations + academic_self_perception
alpha_total, _ = cronbach_alpha(df[all_items])
print("Overall Cronbach's Alpha:", round(alpha_total, 3))


alpha_self, _ = cronbach_alpha(df[academic_self_perception])
print("Academic Self-Perception Alpha:", round(alpha_self, 4))

alpha_exp, _ = cronbach_alpha(df[academic_expectations])
print("Academic Expectations Alpha:", round(alpha_exp, 4))

alpha_work, _ = cronbach_alpha(df[workload_examinations])
print("Workload & Examinations Alpha:", round(alpha_work, 4))
```

*Figure 9: Reliability Analysis using Cronbach's Alpha*

## 6.4 Descriptive Analysis

A custom Python function was developed to calculate frequency counts and percentage distributions for key categorical variables such as gender, academic year, age group, and study environment. This step provides a clear summary of the dataset and helps understand the sample composition before further analysis.

```
def freq_percent(df, column):
    freq = df[column].value_counts().reset_index()
    freq.columns = [column, "Frequency"]
    freq["Percentage (%)"] = (freq["Frequency"] / freq["Frequency"].sum()) * 100
    return freq.round(2)

# Gender
gender_table = freq_percent(df, "gender")

# Academic Year
year_table = freq_percent(df, "academic_year")

# Home Study Environment
home_env_table = freq_percent(df, "home_environment")

# Age Group
age_table = freq_percent(df, "age_group")

# Marital Status
marital_table = freq_percent(df, "marital_status")

# Academic Failure
failure_table = freq_percent(df, "academic_failure")

# Faculty Environment
faculty_env_table = freq_percent(df, "faculty_environment")

# Frequency table
deg_counts = df["degree_program"].value_counts().reset_index()
deg_counts.columns = ["degree_program", "Frequency"]

# Percentage
deg_counts["Percentage (%)"] = (
    deg_counts["Frequency"] / deg_counts["Frequency"].sum() * 100
).round(2)

print(deg_counts)

print("Gender\n", gender_table)
print("\nAcademic Year\n", year_table)
print("\nHome Study Environment\n", home_env_table)
print("\nAge Group\n", age_table)
print("\nMarital Status\n", marital_table)
print("\nAcademic Failure\n", failure_table)
print("\nFaculty Environment\n", faculty_env_table)
```

*Figure 10:Descriptive Analysis*

## 6.5 Inferential Analysis

### Calculation of Overall Stress Score

An overall stress score was calculated by averaging all stress-related variables for each student, creating a single composite measure used as the dependent variable in inferential analysis.

```
# Row-wise mean (one value per student)
df["overall_stress"] = df[all_items].mean(axis=1)
```

*Figure 11: Overall stress score calculation*

### Grouping by Gender

The dataset was split into male and female groups, and their corresponding overall stress scores were extracted for comparison.

```
# Split overall stress by gender
male = df[df["gender"] == "Male"]["overall_stress"]
female = df[df["gender"] == "Female"]["overall_stress"]
```

*Figure 12: Grouping by gender*

### Application of the Welch's t-test

An independent Welch's t-test was conducted to assess differences in mean stress levels between male and female students, as it does not assume equal variances between groups.

```
from scipy.stats import ttest_ind
t_stat, p_value = ttest_ind(male, female, equal_var=False)

print("t-statistic:", round(t_stat, 3))
print("p-value:", format(p_value, ".10f"))
```

*Figure 13:Welch's T test*

### Grouping of Data

The dataset was grouped based on selected categorical variables. Students were categorised by academic year (Year 1–Year 4). For degree programme analysis, Year 3 and Year 4 students were grouped into their respective specialisations, including Finance, Marketing, Human Resource Management, International Business, Accounting, Business Economics, Business Administration, and Hospitality and Leisure Management.

```python
# Create groups
group1 = df[df["academic_year"] == "Year 1"]["overall_stress"]
group2 = df[df["academic_year"] == "Year 2"]["overall_stress"]
group3 = df[df["academic_year"] == "Year 3"]["overall_stress"]
group4 = df[df["academic_year"] == "Year 4"]["overall_stress"]
```

```python
# Step 1: Filter only Year 3 and Year 4
df_deg = df[df["academic_year"].isin(["Year 3", "Year 4"])]

# Step 2: Group by degree programme
groups = [group["overall_stress"].values
          for name, group in df_deg.groupby("degree_program")]
```

*Figure 14: Grouping degree progamme*

## Application of One-Way ANOVA

A one-way ANOVA test was performed to compare the mean stress levels across Academic year and Degree programme. This method evaluates whether there is a statistically significant difference between group means.

```python
# Perform ANOVA
f_stat, p_value = f_oneway(group1, group2, group3, group4)


# Perform ANOVA
f_stat_deg, p_value_deg = f_oneway(*groups)
```

*Figure 15:ANOVA*

## Post-hoc Analysis (Tukey HSD)

When a significant result was obtained, Tukey's Honestly Significant Difference (HSD) test was conducted to identify which specific groups differed from each other [21].

```python
from statsmodels.stats.multicomp import pairwise_tukeyhsd

# Tukey test for Academic Year
tukey = pairwise_tukeyhsd(
    endog=df["overall_stress"],
    groups=df["academic_year"],
    alpha=0.05
)

print(tukey)

from statsmodels.stats.multicomp import pairwise_tukeyhsd

# Run Tukey HSD
tukey_deg = pairwise_tukeyhsd(
    endog=df_deg["overall_stress"],      # dependent variable
    groups=df_deg["degree_program"],     # grouping variable
    alpha=0.05
)

print(tukey_deg)
```

*Figure 16:Tukey Test*

## 6.6 Principal Component Analysis

### i) Data Standardization

The cleaned dataset was obtained, and the data were standardized using StandardScaler to ensure all variables contribute equally to the analysis by removing differences in scale and variance. This is important in PCA and clustering analysis.

```
from sklearn.preprocessing import StandardScaler

# Select only numeric (float/int) columns
numeric_cols = df.select_dtypes(include=['float64', 'int64']).columns

# Apply scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[numeric_cols])
```

*Figure 17: Data standardization*

### ii) Factorability Tests (KMO and Bartlett's Test)

Before performing PCA, dataset suitability was assessed using the KMO test and Bartlett's Test of Sphericity. A KMO value of 0.846 indicates strong sampling adequacy, while a significant Bartlett's test ($p < 0.05$) confirms sufficient correlation among variables, supporting the use of PCA.

```
!pip install factor_analyzer

from factor_analyzer.factor_analyzer import calculate_kmo, calculate_bartlett_sphericity

kmo_all, kmo_model = calculate_kmo(X)
print("KMO:", kmo_model)

chi_square, p_value = calculate_bartlett_sphericity(X)
print("Bartlett p-value:", p_value)
```

*Figure 18:KMO and Bartlett' Test*

### iii) Determination of Number of Factors Using Eigenvalues and Scree Plot

The number of components was determined using eigenvalues and scree plot analysis. Eigenvalues were extracted using the FactorAnalyzer library and plotted against component numbers. The scree plot, as shown in *Figure 19*, a sharp decline followed by a clear levelling off after the fourth component, indicating an elbow point. In addition, the Kaiser criterion (eigenvalue > 1) identified four components with eigenvalues of 4.43, 2.31, 1.39, and 1.07. Based on these criteria, four components were retained for further analysis. Refer to Appendix A (Table 13), for the varimax loadings shows a sharp decline in eigenvalues up to Factor 3, followed by a clear leveling off after Factor 4. This "elbow" indicates that four components should be retained, as additional factors contribute minimal variance.

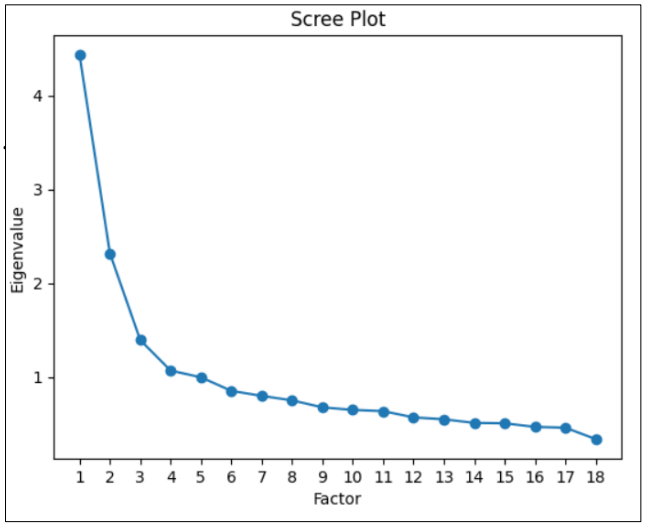


*Figure 19: Scree Plot*

## 6.6 Factor Extraction and Varimax Rotation

Factor analysis was performed using the FactorAnalyzer by specifying four factors based on scree plot and eigenvalue criteria. StandardScaler was applied to standardise the variables, ensuring that all features contribute equally to the PCA by removing differences in scale. Varimax rotation was applied to improve interpretability by ensuring variables load strongly onto specific factors. The resulting factor loadings were organised into a matrix, indicating the strength of relationships between variables and factors, which supported the interpretation and naming of the stress dimensions.

```
fa = FactorAnalyzer(n_factors=3, rotation='varimax')
fa.fit(X_scaled)


import pandas as pd

loadings = pd.DataFrame(
    fa.loadings_,
    index=X.columns,
    columns=["Factor1", "Factor2", "Factor3", "Factor4"]
)

print(loadings)
```

*Figure 20: Factor extraction*

## 6.7 Unsupervised Machine Learning

This section describes the application of unsupervised learning techniques to identify patterns in student stress data without predefined labels. K-Means and Hierarchical clustering were used to group students based on their stress responses and were compared to determine the most

suitable method. Before clustering, the dataset was filtered to include relevant variables, including the stress dimensions (F1–F4), to ensure efficient and focused analysis.

### 6.7.1 K-Means Clustering

Following the extraction of the four principal components (F1–F4), K-Means clustering was applied to group students into distinct segments based on their academic stress profiles. Using the PCA-derived factor scores rather than the original variables reduces dimensionality and multicollinearity, while preserving the most meaningful variation in the data. K-Means Clustering was selected due to its ease, speed, and efficiency in handling larger datasets. The following steps were performed.

i) Data Standardization for Clustering

```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Select only stress dimensions
X = df[['F1', 'F2', 'F3', 'F4']]

# Standardize (VERY IMPORTANT for K-Means)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

*Figure 21: Data Standardization for clustering*

In this step, the four stress dimensions (F1–F4) were selected and standardized using the StandardScaler method. This transformation ensures that all variables are measured on a common scale, preventing bias in distance-based clustering algorithms. As a result, the dataset becomes suitable for applying K-Means clustering to identify distinct student stress profiles.

ii) Determining the Optimal Number of Clusters

Determining the optimal number of clusters (k) is a key step in K-Means. The Elbow Method was used by plotting Within-Cluster Sum of Squares (WCSS) against different values of k. As k increases, WCSS decreases, and the optimal value is identified at the elbow point where the reduction begins to level off. In this study, the elbow was observed at k = 3 as shown in the plot below.

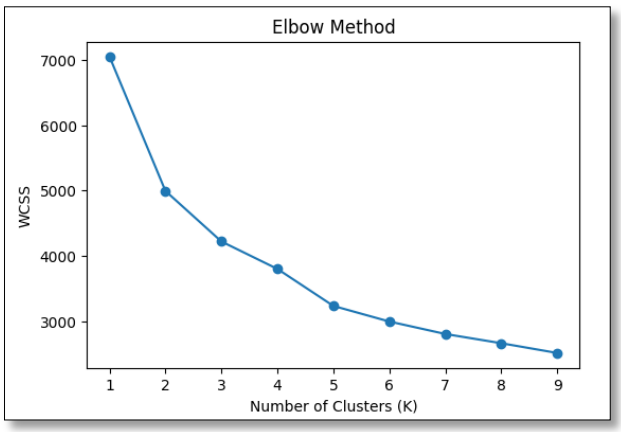


*Figure 22:Elbow Method*

Simultaneously, the Silhouette Score was computed for different values of *k*. A higher score (closer to 1) indicates well-separated clusters, whereas a score near 0 suggests overlapping groupings. The analysis revealed that **k = 2** had the highest average silhouette score compared to other values of *k*.

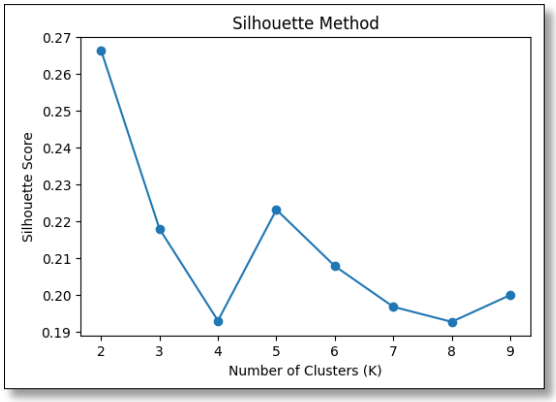


*Figure 23:Sihouette Method*

iii) Evaluating 2 vs 3 Clusters

The Elbow Method indicated that a three-cluster solution (*K = 3*) is appropriate, as it represents the point where the reduction in within-cluster sum of squares (WCSS) begins to level off. However, the Silhouette Score suggested a slightly clearer separation for a two-cluster solution (*K = 2*), indicating a simpler underlying structure in the data.

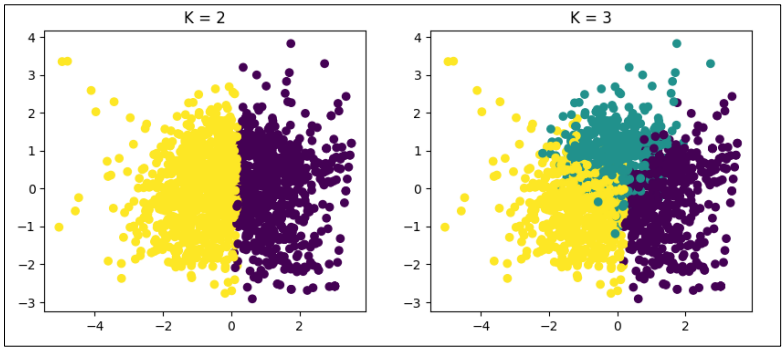


*Figure 24: Cluster Evaluation between 2 and 3*

Relying on a single metric may lead to oversimplified conclusions. Although a two-cluster solution showed better compactness, it risks grouping students into broad categories and masking important differences. As the study aims to identify nuanced stress patterns, a three-cluster solution was preferred. To support this decision, both k = 2 and k = 3 were evaluated using multiple metrics, including the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index with PCA-based visual inspection.

```python
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, davies_bouldin_score, calinski_harabasz_score

results = {}

for k in [2, 3]:
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(X_scaled)

    sil_score = silhouette_score(X_scaled, labels)
    db_score = davies_bouldin_score(X_scaled, labels)
    ch_score = calinski_harabasz_score(X_scaled, labels)

    results[k] = {
        "Silhouette Score": sil_score,
        "Davies-Bouldin Index": db_score,
        "Calinski-Harabasz Index": ch_score
    }

# Print results properly
for k, metrics in results.items():
    print(f"\nK = {k}")
    for name, value in metrics.items():
        print(f"{name}: {value:.3f}")
```

*Figure 25: Validation metrics*

| **Number of Clusters (K)** | **Silhouette Score** | **Davies–Bouldin Index** | **Calinski–Harabasz Index** |
|---|---|---|---|
| 2 | 0.266 | 1.411 | 725.662 |
| 3 | 0.218 | 1.445 | 590.527 |

*Table 2 Cluster Validation Metrics Comparison*

As shown in Table 2, the two-cluster solution performs slightly better across all validation metrics, indicating stronger separation and compactness. However, the differences are minimal and do not significantly reduce clustering quality when using three clusters. PCA visualization also shows that the three-cluster solution maintains reasonable separation. Importantly, the three-cluster model provides better interpretability and aligns with the

conceptual categorization of academic stress, enabling more meaningful student segmentation.

### 6.7.2 Hierarchical Clustering.

Hierarchical Clustering was an alternative unsupervised method that was considered to group students based on similarities in their stress responses. In contrast to K-Means, which needs a fixed number of clusters to be determined in advance. The following steps were performed.

i) Model Development

The Hierarchical Agglomerative Clustering (HAC) approach was applied using the Ward linkage technique, which reduces variance within clusters, producing more meaningful and compact groupings. A dendrogram was generated to visualise how the data points were combined at each step.

```
import scipy.cluster.hierarchy as sch
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 5))

dendrogram = sch.dendrogram(
    sch.linkage(X_scaled, method='ward')
)

plt.title('Dendrogram')
plt.xlabel('Students')
plt.ylabel('Euclidean Distance')
plt.show()
```

*Figure 26: Hierarchical Clustering*

ii) Extracting Cluster Labels

Based on the dendrogram, a 3-cluster solution was chosen to align with the K-Means setup and allow for meaningful comparison. A distance-based cut on the dendrogram was then used to extract cluster labels.

```
from sklearn.cluster import AgglomerativeClustering

hc = AgglomerativeClustering(n_clusters=3, linkage='ward')
df['Hierarchical_Cluster'] = hc.fit_predict(X_scaled)
```

*Figure 27: Extracting Cluster Labels*

iii) Model Evaluation

The resulting HAC cluster labels were assessed using the same clustering metrics as before: Silhouette Score, DBI, and CHI, to determine the compactness and separation of the clusters.

```
from sklearn.metrics import silhouette_score, davies_bouldin_score, calinski_harabasz_score

labels = df['Hierarchical_Cluster']

sil_score = silhouette_score(X_scaled, labels)
db_score = davies_bouldin_score(X_scaled, labels)
ch_score = calinski_harabasz_score(X_scaled, labels)

print("Hierarchical Clustering Evaluation:")
print(f"Silhouette Score: {sil_score:.3f}")
print(f"Davies-Bouldin Index: {db_score:.3f}")
print(f"Calinski-Harabasz Index: {ch_score:.3f}")
```

*Figure 28:Model Evaluation*

K-Means Model Development with Three Clusters

Despite the slightly stronger statistical performance of the two-cluster solution, the three-cluster solution (***K = 3***) was selected as it provides a better balance between statistical validity, interpretability, and the study's objective of identifying diverse student stress profiles.

```
from sklearn.cluster import KMeans

# Apply KMeans
kmeans = KMeans(n_clusters=3, random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

*Figure 29:Model Development with 3 clusters*

4.6.3 Cluster Profiling and Visualisation

To profile each cluster, the mean values of the four stress dimensions (F1–F4) were calculated to identify distinct stress patterns. Radar charts were used to visually compare these patterns across clusters. Additionally, demographic and contextual variables such as gender, academic year, degree programme, home environment, and faculty environment were analysed using frequency and percentage distributions. This combined approach provides a clear understanding of each cluster's characteristics.

## 6.8 Secondary Study

6.8.1 Data Pre-Processing for Prediction

i. Selection of Variables for Predictive Modelling

For the prediction stage, relevant variables were selected, including demographic features (gender, academic failure, faculty environment, home environment, age group, academic year,

and degree programme) and PCA-derived factor scores (F1–F4). Marital status was excluded due to class imbalance. This selection ensures that the model uses both background characteristics and key stress dimensions for prediction.

```
df = df[[
    'gender','marital_status','academic_failure',
    'faculty_environment','home_environment',
    'age_group','academic_year','degree_program',
    'F1','F2','F3','F4'
]]
```

*Figure 30:Variable selection*

**ii.** Simplification of Age Group Categories

The age_group variable was simplified by merging similar categories into a single group. 24–25 and above 26 were recoded into a unified category labeled "24 and above".

This step reduces inconsistencies in categorical data and ensures that closely related age ranges are grouped for more meaningful analysis.

```
df['age_group'] = df['age_group'].replace({
    '24-25': '24 and above',
    'Above 26': '24 and above'
})
```

*Figure 31: Handling class imbalances*

**iv)** Recoding of Faculty Environment Variable

The faculty_environment variable was recoded to combine similar response categories. Specifically, the categories "Dissatisfied" and "Very dissatisfied" were merged into a single category labeled "Not satisfied".

This transformation reduces redundancy in categorical responses and ensures a more consistent representation of dissatisfaction levels.

```
df['faculty_environment'] = df['faculty_environment'].replace({
    'Dissatisfied': 'Not satisfied',
    'Very dissatisfied': 'Not satisfied'
})
```

*Figure 32: Handling class imbalances*

### v) Recoding of Degree Program Variable

The *degree_program* variable was recorded by grouping low-frequency categories (e.g., Business Administration, Hospitality and Leisure Management, and Human Resource Management) into an "Other" category. This was done to address class imbalance, improving data distribution and enhancing the stability and reliability of the predictive model.

```
df['degree_program'] = df['degree_program'].replace({
    'B.B.A. (Hons) in Business Administration': 'Other',
    'B.B.A. (Hons) in Hospitality and Leisure Management': 'Other',
    'B.B.A. (Hons) in Human Resource Management': 'Other'
})
```

*Figure 33: Handling class imbalances*

The *degree_program* variable was cleaned by recoding missing values into a common "General" category. This reflects that Year 1 and Year 2 students are not yet specialised into specific degree programmes.

```
# Replace empty strings with 'General'
df['degree_program'] = df['degree_program'].replace('', 'General')

# Replace actual NaN values with 'General'
df['degree_program'] = df['degree_program'].replace('nan', 'General')
```

*Figure 34: Handling class imbalances*

### vi) Categorical Variable Encoding

Since the predictor variables (X) consisted of categorical features with string values, they were transformed into a numerical format suitable for machine learning. This was achieved using one-hot encoding, which converts each category into binary indicator variables.

```
df_encoded = pd.get_dummies
```

*Figure 35: Categorical variable encoding*

The resulting encoded variables were then converted into integer format to ensure consistency and compatibility with the modelling algorithm. This transformation was necessary because most machine learning models, including regression-based approaches, require numerical input and cannot directly process categorical string data.

```
X_encoded = X_encoded.astype(int)
```

*Figure 36: Encoding*

The fully preprocessed dataset was saved as a CSV file and stored in Google Drive to facilitate efficient access and reproducibility of results. Following this, the predictive modelling process was initiated.

### 6.8.2 Definition of Input and Target Variables for Prediction

The dataset was structured into independent variables (X) and dependent variables (Y) for predictive modelling. The independent variables included demographic features such as gender, academic failure, faculty environment, home environment, age group, academic year, and degree programme. The dependent variables were the PCA-derived factor scores (F1–F4), representing different stress dimensions. This setup enables the prediction of stress profiles based on student characteristics.

```
X = df[[
    'gender','academic_failure','faculty_environment',
    'home_environment','age_group','academic_year','degree_program'
]]

Y = df[['F1','F2','F3','F4']]
```

*Figure 37:Define X and Y variables*

### 6.8.3 Data Splitting for Model Training and Testing

The dataset was divided into training and testing subsets using an 80:20 split. A fixed random state was applied to ensure reproducibility of the results. This approach enables the model to be trained on one portion of the data and evaluated on unseen observations, thereby supporting a more reliable assessment of its generalisation performance.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,    # 80% train, 20% test
    random_state=42   # ensures reproducibility
)
```

*Figure 38: Train and Test*

### 6.8.4 Model Development using XGBoost

The predictive model was developed using the XGBoost regression algorithm, an ensemble method based on gradient boosting. A fixed random_state was used to ensure reproducibility. To predict multiple targets (F1–F4), the model was implemented using a MultiOutputRegressor, enabling simultaneous prediction of all stress dimensions. The model

was trained on the training dataset to learn the relationship between demographic variables and stress dimensions.

```
from xgboost import XGBRegressor
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import r2_score

xgb = XGBRegressor(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=4,
    random_state=42
)

model = MultiOutputRegressor(xgb)

model.fit(X_train, y_train)
```

*Figure 39:XG Boost Model Development*

### 6.8.5 Model Evaluation

After training, predictions were made on the test dataset to evaluate performance on unseen data. Model accuracy was assessed using $R^2$, RMSE, and MAE. $R^2$ indicates the variance explained, while RMSE and MAE measure prediction error, providing an overall evaluation of model performance.

```
y_pred = model.predict(X_test)

r2 = r2_score(y_test, y_pred)
print("R² Score:", r2)

from sklearn.metrics import mean_squared_error, mean_absolute_error
import numpy as np

# RMSE
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

# MAE
mae = mean_absolute_error(y_test, y_pred)

print("RMSE:", rmse)
print("MAE:", mae)
```

*Figure 40: Model Evaluation*

# 7. Results, Analysis, and Discussion

This chapter presents the results obtained from the analysis of academic stress among undergraduate students. The findings are discussed in four main stages: descriptive analysis, inferential statistical analysis, dimensionality reduction using PCA with Varimax rotation, and, as a secondary study, predictive modelling

## 7.1 Descriptive Analysis

### 7.1.1 Demographic Profile of Respondents

i) Gender Distribution

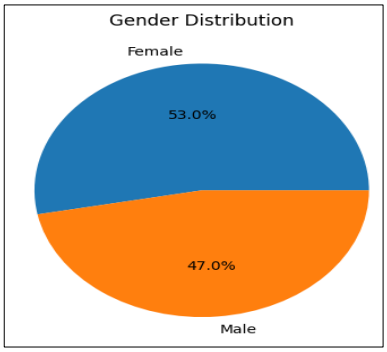


*Figure 41:Gender Distribution*

The sample consists of 935 female students and 828 male students. indicating a relatively balanced gender distribution.

ii) Academic Year Distribution

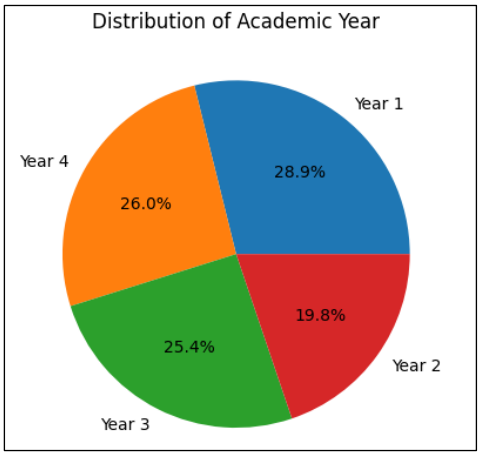


*Figure 42: Academic Year Distribution*

The sample includes 509 students in Year 1, 349 in Year 2, 447 in Year 3, and 458 in Year 4.. The distribution across years is fairly balanced.

vii) Home Study Environment

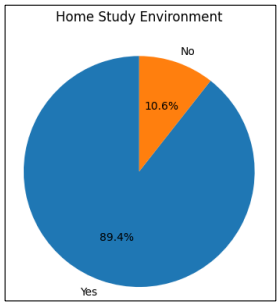


*Figure 43: Home study environment*

Out of the total sample, 1,576 students reported having a home study environment, while 187 reported not having one.

viii) Age Group Distribution

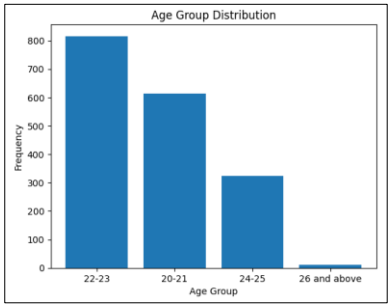


*Figure 44:Age group distribution*

Most respondents fall within the 22–23 age group (46.28%), followed by 20–21 (34.77%), indicating a typical undergraduate age range.High class imbalance between 24-25 and 26 and above.

ix) Marital Status

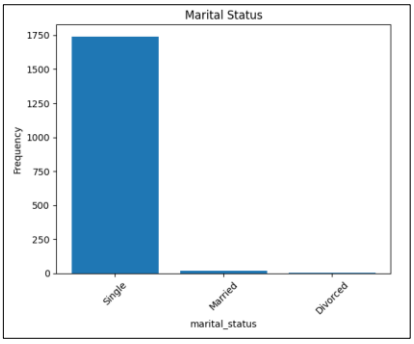


*Figure 45:Marital Status*

The sample is overwhelmingly single (98.70%), which shows a very high class imbalance.

x) Academic Failure

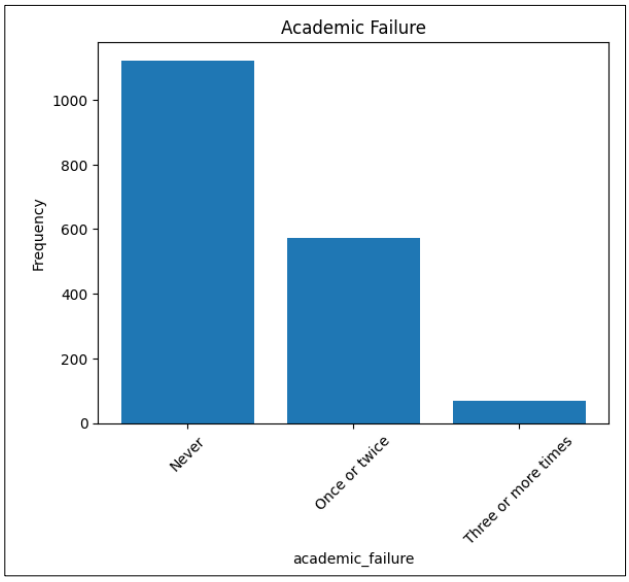


*Figure 46:Academic Failure*

Most students (63.70%) have never experienced academic failure, while a smaller proportion has failed multiple times.

xi) Faculty Environment Satisfaction

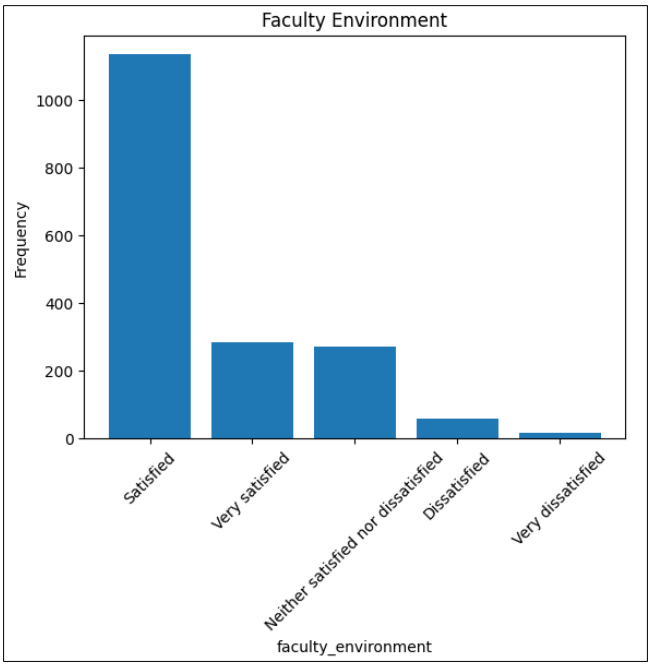


*Figure 47:Faculty Environment Satisfaction*

The majority of students (80.43%) report being satisfied or very satisfied with their faculty environment. High class imbalance between “neither satisfied or dissatisfied”, dissatisfied”, “very dissatisfied”.

xii) Degree Program across year 3 and year 4

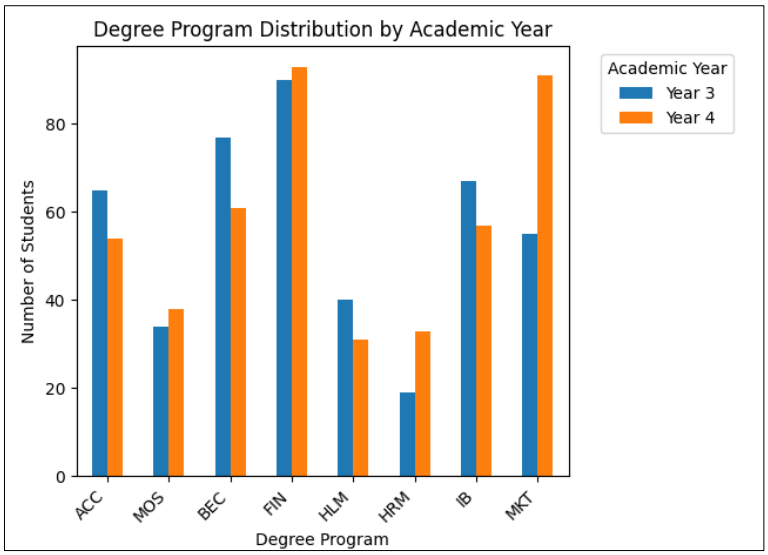


*Figure 48: Degree Programme across year 3 and year 4*

The highest proportion of students is from Finance (20.22%), followed by Marketing and Business Economics.

## 7.2 Inferential Analysis

### 7.2.1 Gender Differences (Welch's t-test)

An independent-samples Welch's t-test was conducted to compare mean stress levels between male and female students. The results indicate a t-statistic of 5.919 and a p-value of 0.0000000039, which is significantly less than the threshold value of 0.05. Since $p < 0.05$, the null hypothesis **($H_0$)** is rejected. There is sufficient statistical evidence to conclude that there is a significant difference in mean stress levels between male and female students.

### 7.2.2 Academic Year Differences (ANOVA) and Post-hoc Analysis

A one-way ANOVA showed significant differences in stress across academic years ($F = 93.728$, $p < 0.001$). Since $p < 0.05$, the null hypothesis is rejected, indicating a significant difference in mean stress levels across academic years. A Tukey HSD post-hoc test showed that all pairwise differences were significant except between Year 1 and Year 2 ($p = 0.3455$). Significant differences were observed between Year 1 and Years 3 and 4, Year 2 and Years 3 and 4, and between Years 3 and 4. Overall, stress levels increase progressively across academic years, with higher-year students experiencing significantly greater stress, including a continued rise from Year 3 to Year 4.

### 7.2.3 Degree Programme Differences (ANOVA)

A one-way ANOVA showed significant differences in stress levels across degree programmes ($F = 11.712$, $p < 0.001$). Since the p-value is less than 0.05, the null hypothesis ($H_0$) is rejected, indicating that mean stress levels differ significantly across degree programmes. Tukey HSD post-hoc results show several significant pairwise differences.

Business Administration students report higher stress than Accounting, Finance, and HRM students, while Business Economics students report higher stress than Accounting students. Hospitality and Leisure Management, International Business, and Marketing students report higher stress than Finance students, with Hospitality and Leisure Management also higher than HRM. However, some comparisons (Accounting vs Finance and Accounting vs HRM) are not significant, suggesting similar stress levels across certain programmes.

## 7.3 PCA and Factor Analysis Results

### 7.3.1 Suitability Tests

Before applying PCA, data suitability was assessed using KMO and Bartlett's test. The KMO value (0.846) indicates adequate sampling adequacy, and Bartlett's test was significant ($p < 0.05$), confirming sufficient correlations among variables. Therefore, the dataset is suitable for PCA.

### 7.3.2 Component Extraction

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset and identify underlying stress dimensions. Based on the Kaiser criterion (eigenvalues > 1), four components were retained. All subsequent components have eigenvalues less than 1 and were not retained. Refer to the eigenvalue table in Appendix A.

The scree plot further supports this decision (*Figure 19)*, which illustrates the distribution of eigenvalues across components. A clear elbow after the fourth component indicates a natural break, with additional components contributing minimal variance. Therefore, four components were selected for further analysis, representing the key underlying dimensions of academic stress among students.

### 7.3.3 Varimax Rotation Results

To enhance interpretability, Varimax rotation was applied to the extracted components. A loading threshold of 0.40 was used to identify significant factor loadings [9]. The rotated component matrix revealed a well-defined four-factor structure, with most variables loading strongly onto a single component while some moderate cross-loadings were observed among examination-related variables. Refer to Appendix A, *Figure 55* for the detailed varimax rotation results and key observations.

### 7.3.4 Identified Stress Dimensions

Based on the rotated component matrix, four key dimensions of academic stress were identified:

1. **Academic Workload and Examination Pressure (F1)**: curriculum_load, assignment_load, difficulty_catching_up, exam_difficulty , exam_time_pressure ,exam_stress

This dimension reflects a cumulative academic pressure process. High curriculum and assignment loads increase the overall academic burden on students, which in turn creates time constraints and difficulty in managing coursework. As students struggle to keep up with academic demands, they perceive examinations as more difficult and experience increased time pressure during assessments. These factors collectively contribute to heightened examination-related stress. Therefore, these variables represent different stages of a continuous process, where excessive academic workload translates into performance pressure and ultimately results in exam-related stress.

2. **Academic and Future Concerns (F2)**: module_failure, exam_concern, employment_concern

Within the university context, unease about examinations can lead to fear of failure, results in delayed graduation due to resits and reductions in GPA. Since academic performance and timely completion of a degree are key determinants of employability, these academic challenges extend into concerns about future career prospects. Therefore, these variables collectively reflect a coherent dimension of stress related to academic uncertainty and its long-term consequences.

3. **Self-Confidence and Personal Management (F3)**: academic_confidence, career_confidence, decision_making, time_management.

This dimension represents students' self-confidence and personal academic capability. It begins with academic confidence, reflecting students' belief in their ability to succeed in their studies. This confidence extends to career confidence, indicating how strongly students believe in their future success beyond university. These beliefs influence students' decision-making ability, where confident students are more capable of making effective academic choices. In turn, strong decision-making supports better time management, enabling students to organise their academic responsibilities efficiently. This dimension functions as a protective factor, where higher confidence supports effective coping and reduces stress, while lower confidence may hinder academic management and contribute to increased stress levels.

**4. Lecturer Expectation and Pressure (F4):** lecturer_strictness, lecturer_expectations

This dimension reflects stress arising from lecturers' behavior and academic expectations**.** It includes students' perceptions that lecturers are strict, frequently point out mistakes, and expect more than students feel capable of achieving. Such high expectations and critical feedback create pressure, leading to increased stress in the academic environment.

### 7.3.5 Overall Interpretation

The four extracted factors collectively explain approximately 51.2% of the total variance, indicating that a substantial proportion of the variability in students' academic stress is captured by the model. This level of explained variance is higher than the 43% reported in the original study [9], suggesting that the factor structure derived in this analysis provides a more comprehensive representation of the underlying data.

The findings confirm that academic stress is a multidimensional construct shaped by several interrelated dimensions reflecting the complex nature of academic stress experienced by UOC FMF students, highlighting context-specific stress patterns influenced by its academic environment.

## 7.4 Cluster Profiling & Insights

Following the implementation of the K-Means clustering algorithm using three clusters, each cluster was analysed according to the derived stress dimensions (F1, F2, F3, F4). This was focused on determining how stress levels varied among different student groups and identifying the key contributors to their stress.

i) **Cluster 0:** High Academic Pressure

Cluster 0 comprises 546 students (53.48% female, 46.52% male) and has the highest stress levels, as shown in *Figure 50*. This cluster shows high scores in Academic Workload and Examination Pressure (F1 = 4.26), Academic and Future Concerns (F2 = 4.15), and Lecturer Expectation and Pressure (F4 = 4.02), along with comparatively low scores in Self-Confidence and Personal Management (F3 = 2.58).

As shown in *Figure 52, t*he majority of students are in Year 4 (36.26%), followed by Year 3 (26.92%) and Year 1 (25.82%), indicating that senior students are more prone to elevated stress, likely due to academic demands and career uncertainty. In terms of degree programmes, the highest contribution is from Finance (28.70%), while the lowest representation is observed in Hospitality and Leisure Management (4.93%), as shown in *Figure 51.*

While most students report a supportive home environment (85.16%), only 55.49% are satisfied with the faculty environment. Overall, this cluster represents highly stressed students, particularly senior undergraduates experiencing strong academic and future-related pressures alongside lower confidence levels.

**Cluster 1 – Moderate Academic Pressure**

Cluster 1 consists of 499 students and is male-dominated (58.52% male, 41.48% female), as shown in *Figure 50*. This cluster exhibits moderate levels of stress across most dimensions, with Academic Workload and Examination Pressure (F1= 3.72) and Academic and Future Concerns (F2 = 3.73).

Notably, it records the lowest level of Lecturer Expectation and Pressure across all three clusters (F4 = 2.24) and the highest level of Self-Confidence and Personal Management across all clusters (F3 = 3.3) among all clusters. These findings suggest that students in this group are better able to manage academic stress, likely due to higher confidence and reduced perceived pressure from lecturers.

The academic year distribution of Cluster 1 is concentrated in Year 3 (33.47%) and Year 4 (32.06%), as shown in *Figure 52*. In terms of degree programmes, International Business (23.24%) dominates, with the lowest contribution coming from Human Resource Management (6.42%)**,** as shown in *Figure 51.*

Additionally, a high proportion of students report a positive home environment (89.18%), and 70.74% express satisfaction with the faculty environment, the highest among all clusters.

Overall, Cluster 1 represents relatively resilient students who experience moderate stress but benefit from higher self-confidence and supportive environments, enabling them to cope more effectively with academic demands.

**Cluster 2 – Low Academic Stress**

Cluster 2 is the largest group, comprising 718 students, and is female-dominated (60.72% female, 39.28% male), as shown in *Figure 50*.

This cluster exhibits the lowest overall stress levels, with relatively lower scores in Academic Workload and Examination Pressure (3.14) and Academic and Future Concerns (F2 = 2.83), while lecturer-related pressure remains moderate (F4 = 2.96).

However, it also records the lowest Lecturer Expectation and Pressure (F3 = 2.40) across all clusters. These findings suggest that although students in this cluster experience lower levels of academic stress, they still face challenges related to confidence and self-management.

The academic year distribution of Cluster 2 is skewed toward early-year students, with the majority in Year 1 (40.95%) followed by Year 2 (26.60%), as shown in *Figure 52*. In terms of degree programme, the highest representation is from Marketing (20.17%)**,** while the lowest contribution is from Human Resource Management (3.43%)**,** as shown in *Figure 51*.

A large proportion of students report a supportive home environment (92.76%), the highest among all clusters, while 66.71% express satisfaction with the faculty environment.

Overall, Cluster 2 represents lower-stress students, primarily in the early stages of their academic journey; however, their comparatively lower confidence levels may indicate potential vulnerability as academic demands increase over time.

**Overall Insight**

Across the four dimensions, clear variations are observed among the clusters. For Academic Workload and Examination Pressure (F1), Cluster 0 records the highest level (4.26), followed by Cluster 1 (3.72), while Cluster 2 shows the lowest (3.14), indicating a decreasing trend in workload-related stress. A similar pattern is seen in Academic and Future Concerns (F2), where

Cluster 0 again has the highest value (4.15), Cluster 1 is moderate (3.73), and Cluster 2 is the lowest (2.83).

In contrast, Self-Confidence and Personal Management (F3) shows a different pattern, with Cluster 1 having the highest level (3.03), followed by Cluster 0 (2.58), while Cluster 2 records the lowest (2.40), indicating that confidence is strongest among moderately stressed students. For Lecturer Expectation and Pressure (F4), Cluster 0 again shows the highest level (4.02), Cluster 2 is moderate (2.96), and Cluster 1 records the lowest (2.24), suggesting that lecturer-related pressure is most intense in the high-stress group and least in the more resilient group.

The clustering results reveal that student stress is not uniform, but varies across three distinct profiles:

- Cluster 0: High stress, low confidence, senior students
- Cluster 1: Moderate stress, high confidence, resilient students
- Cluster 2: Low stress, low confidence, early-year students

```
Mean Stress Levels for Each Cluster:
                F1        F2        F3        F4
Cluster
0         4.256716  4.153236  2.581502  4.017399
1         3.721777  3.728791  3.032565  2.236473
2         3.140204  2.830084  2.400070  2.963788
```

*Figure 49: Mean stress level for each cluster*

This validates the choice of K = 3, as it captures meaningful and actionable differences that would be lost in a simpler two-cluster model.

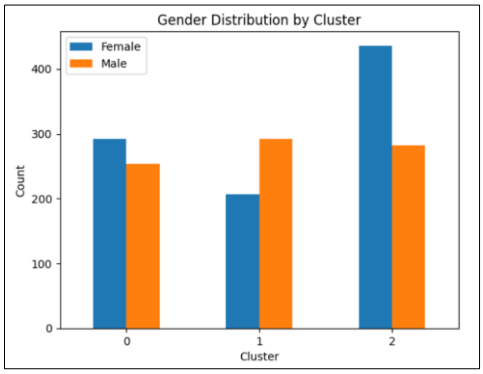


*Figure 51:Gender Distribution by cluster*

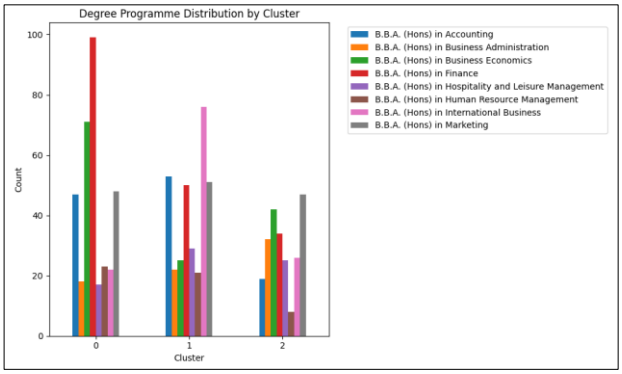


*Figure 50:Degree programme by gender*

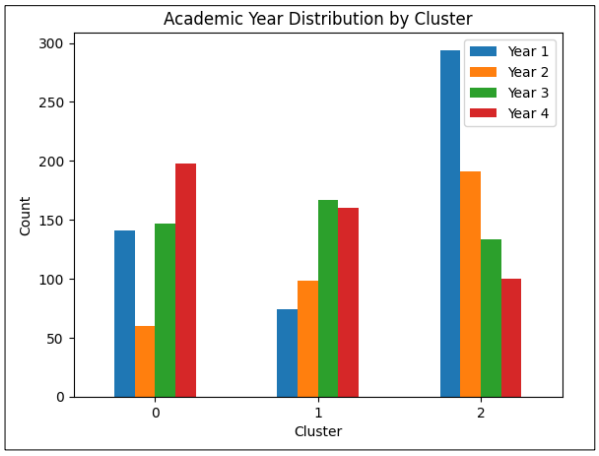


*Figure 52: Academic Year Distribution by cluster*

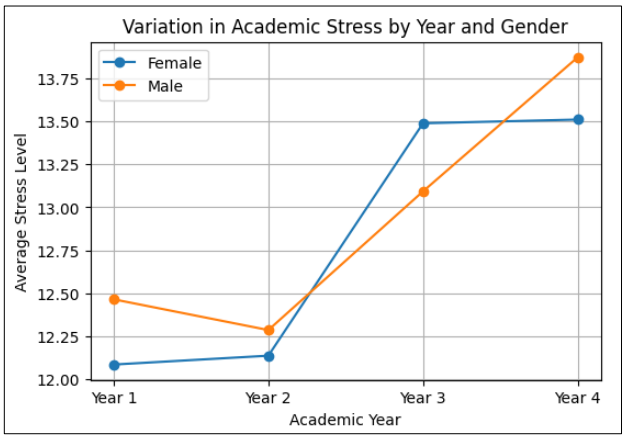


*Figure 53:Variation in academic stress by year and gender*

The line graph (*Figure 53)* indicates that academic stress increases progressively across academic years for both male and female students. In the early years, stress levels remain relatively stable, with male students exhibiting slightly higher stress, possibly due to initial adjustment challenges to university life. However, a sharp increase in stress is observed from Year 3 onwards, particularly among female students, suggesting a critical transition phase characterized by increased academic demands and greater engagement with a complex workload.

As students progress to the final year, stress levels reach their peak, especially among male students. This trend can be attributed to the cumulative effect of multiple academic and personal pressures. Final-year students are required to complete major academic components such as dissertations or capstone projects, which demand substantial time, effort, and independent research skills. At the same time, they face intensified academic workloads and stricter deadlines as they approach graduation.

In addition to academic pressures, uncertainty surrounding future career paths significantly contributes to elevated stress levels. Students in their final year are often engaged in job searches, internship applications, or planning for further studies, all of which introduce additional psychological and emotional pressure. The need to achieve strong academic performance, which may directly influence employment opportunities, further amplifies this stress.

Moreover, final-year students are expected to demonstrate greater independence in their learning, often receiving less direct guidance from lecturers. While this shift toward self-

directed learning is essential for personal and professional development, it may also increase feelings of responsibility and pressure.

These findings are consistent with the cluster analysis, which showed that the high-stress group is predominantly composed of final-year students. Overall, the results highlight that academic stress not only intensifies with progression through university but also varies by gender in terms of its timing and magnitude, driven by a combination of academic workload, future uncertainty, and increased performance expectations.

Based on the findings, several recommendations can be proposed to mitigate academic stress among students. Given that workload pressure emerged as the most significant stressor, universities should consider implementing strategies to manage academic demands better. Attention should be given to improving students' confidence levels, particularly among first-year undergraduates. The analysis indicates that self-confidence is comparatively lower in Year 1 than in other academic years, highlighting a key area of concern. This finding is closely associated with the F3 dimension, which captures aspects related to self-confidence, self-efficacy, and personal academic belief.

Structured support mechanisms such as counselling services, career guidance, and skill development programmes can play a critical role in strengthening students' confidence. In addition, confidence-enhancing initiatives such as mentorship programmes, peer support groups, and orientation workshops can help first-year students better adapt to the university environment.

Furthermore, practical methods such as communication skills training, presentation practice sessions, and academic skill-building workshops can directly improve self-efficacy. Universities can reduce stress levels and support students in developing greater confidence during their transition into higher education.

## 7.5 Interactive Dashboard Visualisation (Power BI)

Following the clustering analysis, the processed dataset was exported and imported into Power BI to develop an interactive dashboard. The dashboard was designed to enhance the visual exploration and interpretation of student stress profiles across different clusters.

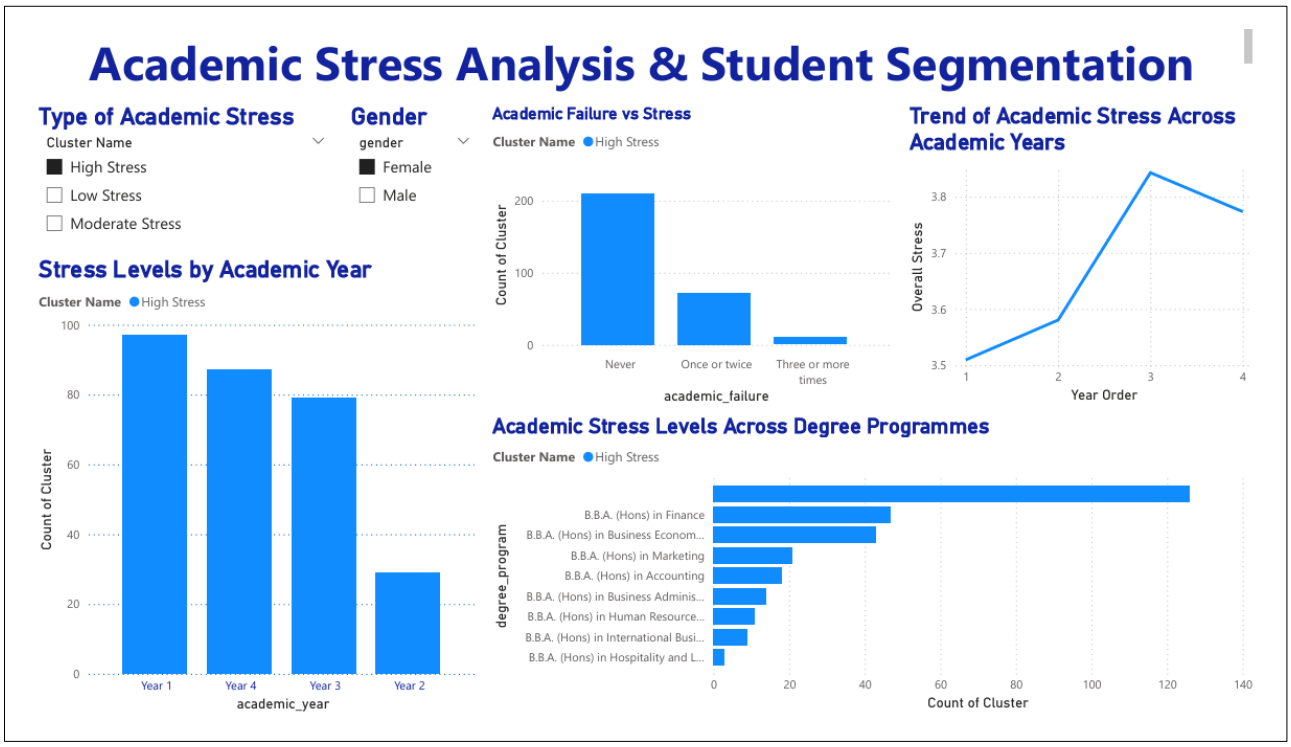


*Figure 54: Dashboard*

## 7.6 Unsupervised Model Performance Comparison

The metrics in Table 4 below evaluate the effectiveness of the unsupervised clustering techniques used to compare the performance of K-Means clustering and Hierarchical clustering for the selected number of clusters ($K = 3$).

*Table 3: Cluster Model Performance Comparison*

| **Model** | **Silhouette Score↑** | **Davies–Bouldin Index** | **Calinski–Harabasz Index** |
|---|---|---|---|
| **K-Means (K=3)** | 0.218 | 1.445 | 590.527 |
| **Hierarchical Clustering (K=3)** | 0.187 | 1.606 | 501.407 |

The Silhouette Score, which measures how well-separated and cohesive the clusters are, is higher for K-Means (0.218) compared to Hierarchical clustering (0.187), indicating better-defined cluster boundaries. Similarly, the Davies–Bouldin Index, where lower values indicate better clustering, is lower for K-Means (1.445) than for Hierarchical clustering (1.606), suggesting that K-Means produces more distinct and less overlapping clusters.

Furthermore, the Calinski–Harabasz Index, which evaluates cluster separation and compactness, is higher for K-Means (590.527) compared to Hierarchical clustering (501.407), reinforcing the superior clustering structure achieved by the K-Means algorithm.

This result suggests that the underlying structure of academic stress patterns among students is better captured by centroid-based partitioning methods such as K-Means, compared to hierarchical approaches. All three evaluation metrics consistently indicate that K-Means clustering outperforms Hierarchical clustering for this dataset.

Therefore, K-Means was selected as the preferred method for segmenting students based on academic stress dimensions.

## 7.7 Predictive Modelling Performance

As an extension to the unsupervised analysis, a predictive modelling approach was implemented to examine the extent to which demographic variables can explain variations in academic stress dimensions.

In this analysis, the demographic variables (gender, academic year, degree programme, home environment, faculty environment, age) were used as input features ($X$), while the stress dimensions derived from Principal Component Analysis (F1–F4) were used as target variables ($Y$).

*Table 4: Predictive Model Performance Metrics*

| **Metric** | **Value** | **Interpretation** |
|---|---|---|
| **$R^2$ Score** | 0.105 | Low explanatory power (10.5% variance explained) |
| **RMSE** | 0.736 | Moderate prediction error |
| **MAE** | 0.573 | The average prediction error is moderate |

As shown in Table 5, the predictive model demonstrates relatively modest performance in explaining variations in academic stress. The $R^2$ score of 0.105 indicates that only approximately 10.5% of the variance in the stress dimensions is explained by the selected demographic variables, suggesting limited explanatory power. Furthermore, the error metrics provide additional insight into model accuracy.

The RMSE value of 0.736 and the MAE of 0.573 indicate a moderate level of prediction error, implying that while the model is capable of identifying general patterns in the data, it lacks precision in accurately predicting individual stress levels.

# 8. Conclusions and Recommendations

## 8.1 Conclusion in Relation to Aim and Objectives

This study successfully achieved its primary aim of conducting a data-driven analysis of academic stress among undergraduate students at UoC FMF. By integrating statistical analysis, dimensionality reduction, clustering, and predictive modelling, the study provides a comprehensive understanding of academic stress aligned with the stated research objectives.

The adoption of the Perception of Academic Stress (PAS) scale ensured the use of a validated and reliable measurement instrument, consistent with existing literature. Through Principal Component Analysis (PCA) with Varimax rotation, four key stress dimensions were identified: academic workload and examination pressure, academic and future concerns, self-confidence and personal management, and lecturer expectation and pressure. These findings confirm that academic stress is multidimensional and influenced by both academic and psychological factors.

## 8.2 Key Findings in Relation to Literature

The findings of this study are consistent with prior research. A key result is the increase in stress levels across academic years, with final-year students experiencing the highest stress. This aligns with literature that highlights increased academic pressure and career uncertainty as major stressors in later stages of study.

Additionally, the study identified variations in stress across degree programmes, suggesting that stress is context-specific and influenced by the nature of academic disciplines. The identification of multiple stress dimensions further supports existing research while contributing new insights specific to the UoC FMF context.

## 8.3 Evaluation and Comparison of Models

The study applied multiple analytical techniques, enabling comparative model evaluation. Principal Component Analysis (PCA) effectively reduced dimensionality while preserving key information and improving interpretability. Varimax rotation further enhanced clarity by simplifying the factor structure, producing distinct loadings that supported meaningful interpretation of the identified stress dimensions.

In clustering, K-Means outperformed hierarchical clustering in terms of cluster separation and clarity. The selection of the three-cluster solution was based not only on validation metrics but also on interpretability in context, demonstrating a balance between statistical accuracy and practical relevance.

The predictive modelling component using XGBoost provided additional insights. While the model is capable of capturing non-linear relationships, its relatively low predictive performance suggests that demographic variables alone are insufficient predictors of academic stress. This highlights the limitation of relying solely on structured data and reinforces findings in the literature that broader psychological and environmental factors influence stress.

## 8.4 Critical Reflection on Findings and Methodology

From a critical perspective, the study demonstrates that academic stress is a complex and context-dependent phenomenon. While advanced analytical techniques were applied, the results indicate that data-driven approaches alone may not fully capture the underlying causes of stress.

The use of PCA simplifies complex data but may overlook subtle variations. Similarly, clustering provides useful segmentation but represents algorithmic groupings rather than absolute categories. The predictive model's limited performance further highlights the importance of including more comprehensive variables.

These observations suggest that while the methodology is effective for identifying patterns, it should be complemented with additional qualitative or psychological insights for a more complete understanding.

## 8.5 Strengths and Limitations

A key strength of this study is the integration of multiple analytical techniques, providing a comprehensive approach to understanding academic stress. The use of a validated scale enhances reliability, while the relatively large sample size improves the robustness of findings within the UoC FMF context. Furthermore, the PAS scale was effectively reduced using PCA with Varimax rotation, which enhanced interpretability by producing a clear and simplified factor structure. This approach allowed for the extraction of meaningful, context-specific stress

dimensions while minimizing redundancy, thereby strengthening the analytical depth and practical relevance of the findings.

However, several limitations exist. The study is restricted to a single faculty, limiting generalisability. The reliance on self-reported data introduces potential bias, and the cross-sectional design does not capture changes over time. Additionally, the limited predictive performance indicates that relevant explanatory variables may not have been adequately captured.

## 8.6 Skills Development and Learning

This project significantly contributed to the development of both technical and analytical skills. Existing knowledge in statistical analysis was strengthened, while new skills were acquired in PCA, clustering, and machine learning techniques such as XGBoost.

The use of Python enhances programming capabilities, particularly in data pre-processing, analysis, and model evaluation. Furthermore, the project improved critical thinking skills, especially in interpreting results, comparing models, and making methodological decisions.

## 8.7 Recommendations and Future Work

Based on the findings, several recommendations can be made. Universities should implement targeted interventions based on student stress profiles. High-stress students may benefit from counselling and workload management support, while students with low confidence may require mentoring and skill development programmes.

Future research should expand the dataset to include multiple faculties or institutions to improve generalisability. Incorporating additional variables, such as psychological factors and academic performance, would enhance predictive modelling. Longitudinal studies could provide deeper insights into how stress evolves. Additionally, developing a decision-support system based on these findings could improve practical application.

## 8.8 Final Conclusion

In conclusion, this study provides a comprehensive and context-specific analysis of academic stress among undergraduate students. Aligning findings with research aims, results, and literature, it demonstrates that academic stress is complex, multidimensional, and influenced by

various factors. The integration of statistical and machine learning techniques enhances understanding, while the findings support the development of targeted interventions to improve student well-being and academic outcomes.

## 9. References

[1] H. Brand and M. Schoonheim-Klein, “Is the OSCE more stressful? Examination anxiety and its consequences in different assessment methods in dental education,” *European Journal of Dental Education*, vol. 13, pp. 147–153, 2009.

[2] V. S. Baste and J. V. Gadkari, “Study of stress, self-esteem and depression in medical students and effect of music on perceived stress,” *Indian Journal of Physiology and Pharmacology*, vol. 58, pp. 298–301, 2014.

[3]B. Rathakrishnan, S. S. B. Singh, A. Yahaya, M. R. Kamaluddin, F. Ibrahim, and Z. A. Rahman, “Academic stress and life satisfaction as social sustainability among university students,” *International Journal of Evaluation and Research in Education (IJERE)*, vol. 11, no. 4, pp. 1778–1786, Dec. 2022, doi: 10.11591/ijere.v11i4.22682

[4] S. C. Lau, H. J. Chow, S. C. Wong, and C. S. Lim, “An empirical study of the influence of individual-related factors on undergraduates’ academic burnout: Malaysian context,” *Journal of Applied Research in Higher Education*, vol. 13, no. 4, pp. 1181–1197, 2020, doi: 10.1108/JARHE-02-2020-0037.

[5]N. Sohail, “Stress and academic performance among medical students,” *Journal of the College of Physicians and Surgeons Pakistan*, vol. 23, pp. 67–71, 2013.

[6] R. Misra and M. McKean, “College students’ academic stress and its relation to their anxiety, time management, and leisure satisfaction,” *American Journal of Health Studies*, vol. 16, pp. 41–51, 2000.

[7] L. Dusselier, B. Dunn, W. Yongyi, M. Shelley II, and D. Whalen, “Personal, health, academic, and environmental predictors of stress in residence halls,” *Journal of American College Health*, vol. 54, pp. 15–24, 2005, doi: 10.3200/JACH.54.1.15-24.

[8] H. Elias, W. S. Ping, and M. C. Abdullah, “Stress and academic achievement among undergraduate students in Universiti Putra Malaysia,” *Procedia - Social and Behavioral Sciences*, vol. 29, pp. 646–655, 2011, doi: 10.1016/j.sbspro.2011.11.288.

[9] D. Bedewy and A. Gabriel, “Examining perceptions of academic stress and its sources among university students: The perception of academic stress scale,” *Health Psychology Open*, vol. 2, pp. 1–9, 2015, doi: 10.1177/2055102915596714.

[10] N. H. Hj Ramli, M. Alavi, S. A. Mehrinezhad, and A. Ahmadi, “Academic stress and self-regulation among university students in Malaysia: Mediator role of mindfulness,” *Behavioral Sciences*, vol. 8, no. 1, p. 12, 2018, doi: 10.3390/bs8010012.

[11]M. C. Pascoe, S. E. Hetrick, and A. G. Parker, “The impact of stress on students in secondary school and higher education,” *International Journal of Adolescence and Youth*, vol. 25, pp. 104–112, 2020, doi: 10.1080/02673843.2019.1596823.

[12] American Psychological Association, *Stress in America™ 2020: A National Mental Health Crisis*. Washington, DC, USA: American Psychological Association, 2020.

[13] B. S. Gamage, N. G. I. A. Deraniyagala, D. Perera, A. A. K. Aishwarya, N. Perera, and L. M. Hettihewa, “Correlation of BMI with educational stress by psychometric analysis via perceived stress scale in non-state university students,” *CINEC Academic Journal*, vol. 6, no. 2, pp. 42–48, 2024, doi: 10.4038/caj.v6i2.124.

[14] G. Liyanage, D. P. R. W. Rajapakshe, D. R. Wijayaratna, J. A. I. P. Jayakody, K. A. M. C. Gunaratne, and A. M. A. D. K. Alagiyawanna, “Psychological distress and sleep quality among Sri Lankan medical students during an economic crisis,” *PLoS ONE*, vol. 19, no. 6, e0304338, 2024, doi: 10.1371/journal.pone.0304338.

[15] G. Barbayannis, M. Bandari, X. Zheng, H. Baquerizo, K. W. Pecor, and X. Ming, “Academic stress and mental well-being in college students: Correlations, affected groups, and COVID-19,” *Frontiers in Psychology*, vol. 13, 2022, doi: 10.3389/fpsyg.2022.886344.

[16] D. Pérez-Jorge, M. Boutaba-Alehyani, A. I. González-Contreras, and I. Pérez-Pérez, “Examining the effects of academic stress on student well-being in higher education,” *Humanities & Social Sciences Communications*, vol. 12, 2025, Art. no. 469. doi: 10.1057/s41599-025-04698-y.

[17] M. M. N. Chathuranga and T. K. N. A. Thalpavila, “Perceived academic stress and coping strategies: Gender-based analysis of management undergraduates of a state university in Sri

Lanka," in *Proceedings of the 16th International Conference on Business and Information (ICBI)*, University of Kelaniya, Sri Lanka, 2025.

[18] W. Rochdiat Murdhiono and V. Vidayanti, "Examining academic stress among co-assistant nursing students using the modified perception of academic stress scale (PAS)," *Indonesian Nursing Journal of Education and Clinic (INJEC)*, vol. 7, no. 1, pp. 1–9, Jun. 2022, doi: 10.24990/injec.v7i1.441.

[19] B. Loera, G. Guidetti, D. Converso, and G. Molinengo, "Assessing the perception of academic students' stress: The psychometric robustness of the PAS scale," *SAGE Open*, vol. 15, no. 4, pp. 1–14, Oct.–Dec. 2025, doi: 10.1177/21582440251393000.

[20] M. A. Khan and M. Saleem, "Validation and adaptation of perception of academic stress scale," *Psychology and Education*, vol. 58, no. 2, pp. 5683–5697, 2021.

[21] R. Beiter, R. Nash, M. McCrady, D. Rhoades, M. Linscomb, M. Clarahan, and S. Sammut, "The prevalence and correlates of depression, anxiety, and stress in a sample of college students," *Journal of Affective Disorders*, vol. 173, pp. 90–96, 2015.

[22] I. J. Ratul, M. M. Nishat, F. Faisal, S. Sultana, and M. A. Mamun, "Analyzing perceived psychological and social stress of university students: A machine learning approach," *Heliyon*, vol. 9, no. 6, p. e17307, 2023. doi:10.1016/j.heliyon.2023.e17307

[23] A. B. Bernardo, E. Blanco, E. Tuero, and J. Herrero, "Academic stress and socio-demographic variables as predictors of university dropout intention," *Electronic Journal of Research in Educational Psychology*, vol. 23, no. 2, pp. 321–348, 2025.

[24] C. H. Liu and T.-C. Hsu, "Using a Hierarchical Clustering Algorithm to Explore the Relationship Between Students' Program Debugging and Learning Performance," in *Proceedings of the 14th International Conference on Learning Analytics & Knowledge (LAK 2024)*, 2024.

[25] S. Sultisiani, A. R. N. Habibi, A. Maulana, H. Tarlongan, A. G. Abo, and A. M. Zaki, "Data Analysis of Student Monitoring Using the K-Means Clustering Method," *Indonesian Journal of Modern Science and Technology (IJMST)*, vol. 1, no. 2, pp. 50–57, May 2025, doi: 10.60421/ijmst.1.2.50-57.2025.

[26] K. Yeler and S. Sürücü, “Classification of Student Stress Levels Using Machine Learning Methods,” *International Journal of Advanced Natural Sciences and Engineering Research*, vol. 9, no. 10, 2025.

[27] M. Razavi et al., “Evaluating Mental Stress Among College Students Using Wearable Sensor Data,” *arXiv preprint arXiv:2309.11097*, 2023.

[28] M. Iatrou and A. Papadopoulos, “Machine Learning vs. Langmuir: A Multioutput XGBoost Regressor Better Captures Soil Phosphorus Adsorption Dynamics,” *Crops*, vol. 5, no. 4, p. 55, 2025, doi: 10.3390/crops5040055.

**[29]** Python Language Reference, Python Software Foundation. [Online]. Available: https://docs.python.org/3/reference/index.html. [Accessed: Mar.18, 2026].

**[30]** W. McKinney, “Data Structures for Statistical Computing in Python,” *Proceedings of the 9th Python in Science Conference (SciPy 2010)*, Austin, TX, USA, Jun. 2010, pp. 56–61. [Online]. Available: https://proceedings.scipy.org/articles/Majora-92bf1922-00a. [Accessed: Mar. 18, 2026].

**[31]** C. R. Harris *et al*., “Array programming with NumPy,” *Nature*, vol. 585, no. 7825, pp. 357–362, Sep. 2020, doi: 10.1038/s41586-020-2649-2.

**[32]** F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

**[33]** T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, San Francisco, CA, USA, 2016, pp. 785–794.

**[34]** P. Virtanen *et al.*, “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, vol. 17, pp. 261–272, 2020.

**[35]** S. Seabold and J. Perktold, “Statsmodels: Econometric and Statistical Modeling with Python,” in *Proc. 9th Python in Science Conf. (SciPy)*, 2010, pp. 57–61.

**[36]** J. D. Hunter, “Matplotlib: A 2D Graphics Environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

**[37]** M. L. Waskom, “Seaborn: Statistical Data Visualization,” *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021.

**[38]** Google, “Google Colaboratory,” 2023. [Online]. Available: https://colab.research.google.com/

## 10. Bibliography

J. F. Hair Jr., W. C. Black, B. J. Babin, and R. E. Anderson, *Multivariate Data Analysis*, 8th ed. Boston, MA, USA: Cengage, 2019.
Book Link

## Appendix A

*Table 5: Description of Demographic and Academic Stress Questions*

| Demographic Variables | | |
|---|---|---|
| **Variable** | **Type** | **Categories / Scale** |
| Gender | Categorical | Male, Female |
| Age Group | Categorical | 20–21, 22–23, 24–25, Above 26 |
| Academic Year | Categorical | Year 1, Year 2, Year 3, Year 4 |
| Marital Status | Categorical | Single, Married, Divorced |
| Degree Programme | Categorical | Finance, Marketing,<br>Human Resource Management,<br>International Business,<br>Accounting,<br>Business Economics,<br>Business Administration,<br>Hospitality and Leisure Management |
| Academic Failure | Ordinal | Never, Once or twice, Three or more times |
| Faculty Learning Environment Satisfaction | Ordinal | Very satisfied, Satisfied, Neither satisfied nor dissatisfied, Dissatisfied, Very dissatisfied |
| Home Study Environment Satisfaction | Categorical (Binary) | Yes, No |
| **Academic Stress Questions** | | |
| **Variable Name (Renamed)** | **Questionnaire Item** | **Data Type** |
| peer_competition | Competition with my peers for grades is quite intense | Ordinal |
| lecturer_strictness | I feel my lecturers are very strict and often point out mistakes in my work | Ordinal |
| lecturer_expectations | My lecturers expect more from me than I can realistically do | Ordinal |
| parental_expectations | The unrealistic expectations of my parents stress me out | Ordinal |
| time_management | The time allocated for lectures, laboratory sessions, and independent study is enough for me to manage my academic workload effectively | Ordinal |
| curriculum_load | The size of the curriculum (workload) is excessive | Ordinal |
| assignment_load | I believe that the amount of work assignments is too much | Ordinal |
| difficulty_catching_up | I am unable to catch up if I fall behind in my work | Ordinal |
| relaxation_time | I have enough time to relax after completing my academic work | Ordinal |
| exam_difficulty | The examination questions are usually difficult | Ordinal |
| exam_time_pressure | The allocated examination duration is too short for me to complete my answers properly | Ordinal |
| exam_stress | Examination periods are very stressful to me | Ordinal |

| | | |
|---|---|---|
| academic_confidence | I am confident that I will be a successful student | Ordinal |
| career_confidence | I am confident that I will be successful in my future career | Ordinal |
| decision_making | I find it easy to make academic decisions | Ordinal |
| module_failure | I fear failing courses/modules this year | Ordinal |
| exam_concern | I think that my worry about examinations is a weakness of character | Ordinal |
| employment_concern | Even if I pass my exams, I am worried about getting a job | Ordinal |

*Table 6: Reverse-Coded Variables*

| **Original Question (Main Item)** | **Renamed Variable** |
|---|---|
| Competition with my peers for grades is quite intense | peer_competition |
| I feel my lecturers are very strict and often point out mistakes in my work | lecturer_strictness |
| My lecturers expect more from me than I can realistically do | lecturer_expectations |
| The unrealistic expectations of my parents stress me out | parental_expectations |
| The size of the curriculum (workload) is excessive | curriculum_load |
| I believe that the amount of work assignments is too much | assignment_load |
| I am unable to catch up if I fall behind in my work | difficulty_catching_up |
| The examination questions are usually difficult | exam_difficulty |
| The allocated examination duration is too short for me to complete my answers properly | exam_time_pressure |
| Examination periods are very stressful for me | exam_stress |
| I fear failing courses/modules this year | module_failure |
| I think that my worry about examinations is a weakness of character | exam_concern |
| Even if I pass my exams, I am worried about getting a job | employment_concern |

*Table 7 Eigenvalues*

| Factor | Eigenvalue |
|---|---|
| 1 | 4.434974 |
| 2 | 2.318979 |
| 3 | 1.394768 |
| 4 | 1.069456 |
| 5 | 0.997734 |
| 6 | 0.853363 |
| 7 | 0.799706 |
| 8 | 0.752306 |
| 9 | 0.677626 |
| 10 | 0.65059 |
| 11 | 0.637368 |
| 12 | 0.570994 |
| 13 | 0.550745 |
| 14 | 0.512439 |
| 15 | 0.507953 |
| 16 | 0.469631 |
| 17 | 0.461371 |
| 18 | 0.339998 |

*Table 8: Rotated Component Matrix for Academic Stress Factors*

| Item | Factor 1 | Factor 2 | Factor 3 | Factor 4 |
|---|---|---|---|---|
| Curriculum load | 0.53 | | | |
| Assignment load | 0.61 | | | |
| Difficulty catching up | 0.47 | | | |
| Exam difficulty | 0.50 | | | |
| Exam time pressure | 0.47 | | | |
| Exam stress | 0.47 | | | |
| Module failure | | 0.58 | | |
| Exam concern | | 0.50 | | |
| Employment concern | | 0.62 | | |
| Academic confidence | | | 0.75 | |
| Career confidence | | | 0.80 | |
| Decision making | | | 0.52 | |
| Time management | | | 0.44 | |
| Lecturer strictness | | | | 0.64 |
| Lecturer expectations | | | | 0.62 |

| | Factor1 | Factor2 | Factor3 | Factor4 |
|---|---|---|---|---|
| peer_competition | NaN | NaN | NaN | NaN |
| lecturer_strictness | NaN | NaN | NaN | 0.643675 |
| lecturer_expectations | NaN | NaN | NaN | 0.624204 |
| parental_expectations | NaN | NaN | NaN | NaN |
| time_management | NaN | NaN | 0.442909 | NaN |
| curriculum_load | 0.533196 | NaN | NaN | NaN |
| assignment_load | 0.606369 | NaN | NaN | NaN |
| difficulty_catching_up | 0.468033 | NaN | NaN | NaN |
| relaxation_time | NaN | NaN | NaN | NaN |
| exam_difficulty | 0.503719 | NaN | NaN | NaN |
| exam_time_pressure | 0.466970 | 0.412383 | NaN | NaN |
| exam_stress | 0.469033 | 0.458225 | NaN | NaN |
| academic_confidence | NaN | NaN | 0.752109 | NaN |
| career_confidence | NaN | NaN | 0.804239 | NaN |
| decision_making | NaN | NaN | 0.521186 | NaN |
| module_failure | NaN | 0.580372 | NaN | NaN |
| exam_concern | NaN | 0.497974 | NaN | NaN |
| employment_concern | NaN | 0.619454 | NaN | NaN |

*Figure 55:Factor Loadings Table*

Key observations include:

The rotated component matrix obtained through PCA with Varimax rotation provides a detailed representation of the underlying structure of academic stress by illustrating the strength of relationships between variables and the extracted factors. In line with established practice and prior studies [9], a factor loading threshold of 0.40 was applied to identify significant variable–factor associations, with values below this threshold treated as non-significant (represented as NaN), thereby enhancing interpretability by retaining only meaningful contributions.

Factor 1 shows high loadings for variables such as assignment load (0.606), curriculum load (0.533), exam difficulty (0.504), and exam stress (0.469), along with difficulty catching up and exam time pressure, indicating a strong dimension of academic workload and examination pressure.

Factor 2 is dominated by employment concern (0.619), module failure (0.580), and exam concern (0.498), with additional contributions from exam-related variables, reflecting academic and future-related anxieties.

Factor 3 includes strong loadings for career confidence (0.804), academic confidence (0.752), decision making (0.521), and time management, representing a well-defined dimension of self-confidence and personal capability.

Factor 4 is primarily associated with lecturer strictness (0.644) and lecturer expectations (0.624), capturing lecturer-induced or external academic pressure.

Some variables, particularly exam time pressure and exam stress, exhibit cross-loadings across Factors 1 and 2, indicating that these variables are linked to both workload-related and concern-based dimensions of stress. In such cases, variables were assigned to the factor on which they had the highest loading to maintain conceptual clarity and consistency with the dominant interpretation of each factor.

Overall, the factor structure is clear and well-defined, with most variables loading strongly onto a single factor, demonstrating the effectiveness of Varimax rotation in producing a simplified and interpretable structure. The presence of limited cross-loadings further highlights the inherently multidimensional and interconnected nature of academic stress.

## Appendix B

Ethics Permission Letter

Survey Questionnaire (Google Form Link)

GitHub Repository

Responses Link

Supervisor Log Report

GITHUB-Repository

Model Explanation Video